



DRAW MIND AI

Supervisors: Dr. RUBA OBEIDAT, Dr. DIMA SULEIMAN

STUDENT NAME	Registration NO.
RAHMEH IBRAHIM ALFUQAHAA	2211873
RAGHAD Wael ALKHUdAIR	0228350
YARAH ALHINDI	2213245
MARIYA KHALED HAMMASH	0228302

**KING ABDULLAH II SCHOOL OF INFORMATION TECHNOLOGY
DATA SCIENCE AND ARTIFICIAL INTELLIGENCE
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1. INTRODUCTION:

1.1 Project Motivation:

Understanding a child's psychological state through conventional verbal communication presents a persistent challenge in child psychology, particularly for children in early developmental stages who have not yet acquired the linguistic capacity to articulate complex emotional experiences. This limitation creates a significant gap between what a child feels and what a clinician or caregiver can reliably observe. Children, however, frequently externalize internal emotional states through non-verbal channels, most notably through free-hand drawing. Drawing is a spontaneous and uninhibited medium through which children represent their perception of the world, their fears, and their sense of wellbeing without the social pressure inherent in verbal expression.

Despite the well-documented relationship between children's drawings and their psychological states, the systematic analysis of such drawings remains a largely manual and highly subjective process. Practitioners rely on personal experience and pattern recognition developed over years of clinical work, which introduces inter-analyst variability and limits the scalability of drawing-based assessment to clinical settings alone. The absence of a consistent, technology-assisted framework means that parents, teachers, and non-specialist caregivers are effectively excluded from using this valuable observational channel.

This project was motivated by the opportunity to bridge this gap through the application of deep learning and explainable artificial intelligence. By developing a system capable of classifying the emotional content of children's drawings across four categories — happiness, sadness, anger, and fear — and providing interpretable explanations for each prediction, the project aims to transform an expert-only activity into an accessible, transparent, and scalable tool. The intent is not to automate clinical judgment, but to provide a preliminary analytical layer that enriches the observational capacity of parents and educators while preserving the essential role of trained professionals.

1.2 Problem Statement:

The accurate assessment of a child's emotional state is a prerequisite for effective psychological support, yet traditional assessment methods depend heavily on verbal self-reporting, which is developmentally inappropriate for young children. Children between the ages of three and ten years lack the metacognitive and linguistic abilities required to describe internal emotional experiences with the specificity needed for clinical assessment. This creates a systematic blind spot in early childhood mental health screening.

While drawing-based assessment techniques have been employed in clinical psychology for decades, their implementation suffers from three interrelated problems. First, the interpretation of drawings is non-standardized and varies considerably between practitioners, reducing the reliability of assessments across different clinical contexts. Second, the process is time-intensive and requires

trained clinical staff, making it unsuitable for routine monitoring in educational or home settings. Third, existing computational approaches to drawing analysis have focused primarily on adult art or generalized image classification, with very limited datasets and models specifically designed for children's emotional drawings.

A particular challenge specific to children's emotional drawing datasets is the natural imbalance in the representation of emotional categories. Drawings expressing happiness are significantly more common than those expressing anger or fear, which are rare emotional expressions that children produce less frequently and that are less often collected by researchers. This class imbalance makes standard binary or multi-class classification architectures unreliable when applied to the full emotional spectrum. No existing open-source dataset adequately represents all four emotional categories with sufficient volume and diversity for robust model training.

This project addresses these challenges through the design and implementation of a hierarchical fusion classification system that leverages two specialized binary models — one trained on happiness and sadness, and one trained on anger and fear — whose outputs are combined by a learned merger network. This architecture directly addresses the dataset scarcity and imbalance problem while producing a four-class emotional classification that is both accurate and interpretable.

1.3 Project aim and Objectives:

- **Project Aim:**

The primary aim of this project was to design, train, and evaluate an AI-powered system for the preliminary emotional analysis of children's hand-drawn artwork, producing interpretable output accessible to non-specialist users.

The specific objectives pursued in the realization of this aim were as follows:

1. To construct and curate a domain-specific multi-source dataset for children's emotional drawing classification, addressing the class imbalance inherent in publicly available data through strategic dataset merging, deduplication, and targeted augmentation.
2. To design and train two specialized binary classification models using MobileNetV2 with a custom SimAM attention mechanism: one model targeting the happiness/sadness dimension and one targeting the anger/fear dimension.
3. To develop a hierarchical feature-level fusion architecture that concatenates the embedding representations of both binary models and trains a merger network to produce a unified four-class emotional prediction.
4. To implement a multi-modal explainability framework incorporating Gradient-weighted Class Activation Mapping (Grad-CAM) and Visual features of the image, enabling visual attribution of model predictions.

5. To integrate a large language model (Grok) as an interpretation layer that translates raw model outputs and visual features into natural-language reports suitable for parents and educators.

6.1. To evaluate the full system using standard classification metrics including accuracy, precision, recall, and F1-score across all four emotional categories, and to validate predictions using Test-Time Augmentation (TTA).

- Project Scope:

This project encompassed the full pipeline from raw data collection to model deployment within a mobile application interface. The technical scope included two independent binary classification models, a learned merger network combining their outputs, an explainable AI layer, a large language model integration, and a mobile-accessible user interface designed for parents, teachers, and caregivers.

The system was designed to accept images of hand-drawn children's artwork and return a four-class emotional classification accompanied by visual attribution maps and a natural-language interpretation report. The analysis is strictly limited to visual features of the drawing as supplemented by optional parent or caregiver text notes. The system does not provide clinical diagnoses, psychiatric assessments, or any form of medical recommendation. All outputs are explicitly framed as preliminary analytical indicators intended to support — and not replace — professional psychological evaluation.

The scope excludes real-time drawing capture for analysis, longitudinal automated monitoring, analysis of digital drawings produced on touch devices without stylus, and any form of biometric or facial expression analysis. The classification framework covers four emotion categories: happiness, sadness, anger, and fear. Compound or mixed emotional states are not classified as discrete categories by the model, though the language model interpretation layer may acknowledge emotional ambiguity in its generated reports.

1.4 Project Software and Hardware Requirements:

A. Software Requirements:

Item	Requirements
Operating System	Android / iOS
Camera Permission	capture images directly from the App
Language	English language support (Arabic can be added)
Operating System	Windows 11
Programming Language	Python + React
Back-End Framework	FastAPI
Artificial Intelligence Technologies	TensorFlow
Image Processing	OpenCV
Database	Postgres SQL
Security	Data encryption and user privacy protection

Table 1: Software Requirements

B. Hardware Requirements:

Component	Requirements
Device	Tablet or Smartphone
Processor (CPU)	Dual-core processor or higher
Memory (RAM)	Minimum 2 GB (4 GB recommended)
Display	Screen resolution of at least 720p
Drawing Tools	Touch screen (digital pen is optional if available)
Camera	Built-in camera to capture the drawing image
Internet Connection	Stable internet connection (Wi-Fi or mobile data)
Storage	Images are stored within the application
Processor (CPU)	Multi-core processor (4 cores or more)
Memory (RAM)	Minimum 16 GB
Graphics Processing Unit (GPU)	Graphics card to support image analysis (optional but recommended) + for running the models
Storage	SSD with a minimum capacity of 250 GB
Network	High-speed network connection
Backup	Data backup system
Power	Uninterruptible Power Supply (UPS)

Table 2: Hardware Requirement

1.5 Project Limitations:

The system operates under several constraints that **define the boundaries of its reliability and applicability**:

Image Quality Sensitivity: Model performance is sensitive to image quality. Photographs of drawings taken in poor lighting, at oblique angles, or with significant motion blur may reduce classification accuracy, as the preprocessing pipeline is optimized for reasonably clear, front-facing captures.

Cultural and Stylistic Bias: The training datasets, while curated from multiple sources and augmented, remain limited in cultural diversity. The emotional **expression conventions** in children's drawings vary across cultural backgrounds, and the system's generalizability to drawing styles significantly different from those represented in the training data has not been fully validated.

Dataset Size Constraints: The core models were trained on datasets that, even after targeted aggregation and cryptographic deduplication, are relatively small compared to large-scale computer vision benchmarks. This constrains the upper bound of achievable accuracy and makes the model susceptible to overfitting on underrepresented visual patterns, particularly in the high-arousal (Anger/Fear) categories.

Connectivity Dependency: The system requires an active internet connection to query the Grok LLM API for contextual synthesis and natural-language report generation. Offline environments are only supported for the raw visual classification phase, not the full analytical pipeline.

Cascading Logic Errors: The four-class hierarchical system depends on the accurate performance of the foundational base models. Because the system relies on a sequential decision-logic module, misclassifications in the initial valence phase (e.g., failing to flag a drawing as 'Sad') will prevent the secondary arousal model from running, meaning errors can cascade through the thresholds and compound in highly ambiguous cases.

1.6 Project Expected Output:

The project produced a functioning multi-component AI system comprising two trained binary classification models, a hierarchical decision-logic module, an explainability module, and a language model integration layer. The complete pipeline accepts an image of a child's drawing

and produces:

- (a) a four-class emotional classification with confidence scores, (b) Grad-CAM visual attribution heatmaps identifying the drawing regions most influential to the prediction, (c) a structured natural-language report describing the emotional indicators detected, and (d) practical suggestions for the parent or caregiver based on the observed emotional context.

These outputs are delivered through a mobile application interface that supports profile management for multiple children, historical tracking of emotional analyses over time, and optional text input from parents or teachers that contextualizes the automated analysis. The system is presented as a preliminary analytical tool; all outputs include explicit disclaimers recommending professional consultation in cases where emotional concern is identified.

1.7 Project Schedule:

Task #	Task Name	Dependencies
T #1	Gather Project Requirements	None
T #2	Project Planning	T #1
T #3	Design Database Architecture	T #1, T #2
T #4	Design AI Architecture (Cascading CNNs & LLM API)	T #1, T #2
T #5	Develop and Deploy the Database	T #3
T #6	Gather Image Dataset (Children's Drawings)	T #1, T #4
T #7	Image Data Preprocessing & Augmentation	T #6
T #8	Train Tiered CNN Models (Happy/Sad & Angry/Fear)	T #7

T #9	Integrate & Prompt Engineer External LLM (Groq API)	T #4
T #10	Develop FastAPI Backend & AI Pipeline	T #5, T #8, T #9
T #11	Design User Interface (Frontend)	T #1
T #12	System Integration & Local Deployment	T #10, T #11
T #13	System Testing & Final Documentation	T #12

Table 3: Project Schedule

1.10 Report Outline:

This report is organized as follows: Chapter Two presents the theoretical background, a literature review of deep learning approaches, and an analysis of related existing systems. Chapter Three details the system requirements engineering, encompassing feasibility studies, functional requirements, the dataset construction strategy, and the custom master preprocessing filter. Chapter Four describes the system design, covering architectural modeling through DFD and UML diagrams, interface prototypes, and the overarching AI system design. Chapter Five documents the system implementation, detailing the binary base models, the hierarchical architecture, explainable AI (XAI) integration, backend development, and the Large Language Model (LLM) prompt engineering. Chapter Six presents comprehensive system testing and evaluation, bridging software quality validation with detailed experimental results for the AI models. Finally, Chapter Seven states the project conclusions, outlines system strengths and weaknesses, and proposes directions for future work.

2. BACKGROUND, LITERATURE REVIEW, AND RELATED EXISTING SYSTEMS:

2.1 Introduction:

This chapter outlines the core architecture, data requirements, and technical methodology behind the DrawMind AI platform. To build a reliable tool for analyzing children's drawings and text inputs, the system must bridge the gap between computer vision and child psychology. This requires a carefully structured workflow capable of processing multi-modal inputs—specifically, visual drawing features and contextual descriptions from parents or children.

The following sections break down the complete engineering lifecycle of the project. We start by outlining the project's timeline, task dependencies, and core system requirements. From there, we detail our data collection process, data preprocessing pipelines, and the experimental choices made while training our two-tiered cascading CNN models. Finally, we discuss how these local vision models interface with external LLM APIs to generate comprehensive, psychology-grounded analytical reports for end-users.

2.2 Theoretical Background:

2.2.1 Children's Drawings as Psychological Indicators:

Children's drawings have long been considered valuable tools for understanding children's emotional and psychological states. Since children may experience difficulty expressing their feelings verbally, drawings provide a nonverbal communication method that reflects their internal emotions, perceptions, and experiences. According to psychological theories, visual elements

such as color usage, object placement, line intensity, spatial organization, and drawing size may indicate emotions including happiness, sadness, anxiety, fear, or emotional distress.

Several psychological assessment methods have been developed based on children's drawings, including the Draw-a-Person test proposed by Goodenough and the House-Tree-Person (HTP) test introduced by Buck. These approaches are widely used in psychology to evaluate emotional development, cognitive abilities, self-perception, and behavioral conditions. Researchers have also used children's drawings to investigate trauma, bullying, autism, depression, and emotional well-being.

Traditional drawing analysis mainly depends on professional psychologists and counselors, making the process subjective, time-consuming, and costly. Different specialists may interpret the same drawing differently depending on personal experience and judgment. Recent advances in artificial intelligence and computer vision have introduced computational approaches capable of extracting visual features automatically from children's drawings. These approaches allow more objective and scalable emotional assessment by analyzing hidden visual patterns that may not be noticeable through manual observation alone.

Recent research by Serdar Çiftçi and colleagues introduced the KIDO dataset (Çiftçi, Koçakoğlu, Karci, Karaca, & Söylemez, 2025), which contains scanned children's drawings paired with written reflections describing emotions such as happiness and sadness. Their study demonstrated that deep learning models can successfully analyze emotional patterns in children's drawings and assist in early emotional assessment and psychological support application.

2.2.2 Emotion Classification in Computer Vision:

Emotion classification in computer vision refers to the automatic recognition of emotional states from visual data using artificial intelligence and deep learning techniques. Recent developments in computer vision have enabled machines to analyze images and detect meaningful emotional patterns through feature extraction and image classification algorithms.

In the context of children's drawings, computer vision models can analyze visual characteristics such as color distribution, stroke thickness, texture, spatial arrangements, and object composition to identify emotional indicators. Deep learning techniques have significantly improved image classification performance because they automatically learn hierarchical visual representations directly from data without requiring handcrafted feature engineering.

Convolutional Neural Networks (CNNs) and Vision Transformer (ViT)-based architectures are among the most widely used models for visual emotion recognition tasks. Studies have shown that these models can classify emotions such as happiness and sadness from children's drawings with promising accuracy levels. In the KIDO study, CNN-based and transformer-based models achieved high performance in emotion recognition tasks, demonstrating the effectiveness of deep learning for analyzing children's emotional expressions through drawings.

Computer vision has also been applied in related psychological applications, including autism detection, trauma identification, ADHD diagnosis, and behavioral analysis using children's drawings and handwriting patterns. These applications highlight the growing role of AI technologies in supporting psychological and educational assessment systems.

2.2.3 Transfer Learning and Convolutional Neural Networks:

Convolutional Neural Networks (CNNs) are deep learning architectures specifically designed for image analysis and visual recognition tasks. CNNs consist of convolutional, pooling, and fully connected layers that enable automatic extraction of hierarchical visual features such as edges, textures, shapes, and complex patterns. Compared to traditional machine learning approaches that rely on handcrafted features, CNNs learn feature representations directly from training data, resulting in improved performance in image classification problems.

Training deep neural networks from scratch requires large datasets and high computational resources. To overcome this limitation, transfer learning has become a widely adopted approach in computer vision applications. Transfer learning utilizes pretrained models that were previously trained on large-scale datasets such as ImageNet and adapts them to specific tasks through fine-tuning.

Several pretrained architectures have demonstrated strong performance in image classification tasks, including VGG19, ResNet101, DenseNet121, EfficientNet, ConvNeXt, and MobileNetV2. These models provide efficient feature extraction capabilities while reducing training time and improving classification accuracy. In the KIDO study, pretrained CNN and Vision Transformer models were fine-tuned to classify children's emotions, educational levels, and gender based on their drawings.

Transfer learning enables models to recognize subtle emotional features such as stroke intensity, drawing complexity, object structure, and color usage even when limited training data is available. This makes transfer learning highly suitable for psychological image analysis applications where collecting large-scale labeled datasets may be difficult.

2.2.4 Explainable AI in Medical and Psychological Applications:

Explainable Artificial Intelligence (XAI) refers to techniques and methods designed to make artificial intelligence systems more transparent and interpretable to humans. In sensitive domains such as healthcare and psychology, explainability is essential because AI systems may influence decisions related to emotional assessment, diagnosis, and mental health support.

Deep learning models are often considered black-box systems because their internal decision-making processes are difficult to interpret. This lack of transparency may reduce trust in AI-generated predictions, especially in psychological applications involving children. Explainable AI

techniques address this issue by identifying and visualizing the features or image regions that contribute most to model predictions.

Several interpretability methods have been applied in computer vision systems, including saliency maps, occlusion analysis, integrated gradients, Grad-CAM, and attention visualization techniques. These approaches help researchers and professionals understand how AI models classify emotional states from visual inputs.

The KIDO study applied explainability techniques using the Captum framework to visualize which regions of children's drawings influenced the model's predictions. Methods such as integrated gradients, saliency maps, and occlusion analysis were used to highlight emotionally significant areas including dark strokes, object placement, and color concentration patterns. These visualization techniques improve the reliability, transparency, and ethical use of AI systems in psychological and educational environments.

2.3 Literature Review:

2.3.1 Emotion Recognition from Children's Drawings:

In child psychology, drawing-based assessments like the House-Tree-Person (HTP) test have been around for a long time. The basic idea is that the way a child draws the lines they use, the colors they choose, how much space they fill can reveal things about their emotional state that they might not be able to put into words. Traditionally though, interpreting these drawings has always been a manual process, relying on a trained psychologist to look at the drawing and make a judgment call. The problem with that approach is that it's inherently subjective. Two psychologists looking at the same drawing might reach different conclusions. That's where computational methods come in. By converting the visual properties of a drawing into numerical features things like edge density, color distribution, and spatial usage it becomes possible to analyze those variables consistently and objectively, without the variability that comes with human judgment. This shift from qualitative interpretation to structured mathematical analysis is what makes automated emotion recognition from children's drawings a genuinely useful research direction, rather than just a technical novelty.

2.3.1 Deep Learning Approaches for Drawing and Sketch Classification:

Early attempts at classifying hand-drawn artwork using machine learning leaned heavily on hand-crafted feature descriptors like HOG and SIFT. These techniques work by extracting specific spatial patterns from an image — edge orientations, keypoints, that kind of thing — and they perform reasonably well when the drawings follow predictable structures. The problem is that children's drawings don't really do that. They're messy, abstract, and highly variable from one child to the next, and static feature descriptors just aren't flexible enough to handle that kind of unpredictability. They tend to miss the subtler details irregular stroke patterns, unusual compositions, non-standard proportions that are actually meaningful in a psychological context. CNNs changed things quite a bit. Instead of relying on manually designed features, CNNs learn their own feature representations directly from the training data, building up from low-level edges and textures in the early layers to more abstract, high-level patterns deeper in the network. This hierarchical learning process is a much better fit for the kind of variation found in children's drawings.

More recently, transfer learning has made this even more practical. Rather than training a deep network from scratch which requires a lot of data and compute you can take a model that was already trained on a large image dataset and fine-tune it for your specific task. MobileNetV2 is a particularly good choice for this kind of project because its depthwise separable convolutions significantly reduce the number of floating-point operations and overall model size, making it efficient enough to run on mobile devices without sacrificing too much in terms of what the model can actually learn. Despite being lightweight, it still produces a rich feature representation that can capture fine-grained details like stroke texture and line variation exactly the kinds of visual cues that matter here.

2.3.2 Attention Mechanisms in Image Classification:

One issue with standard convolutional layers is that they treat every part of an image the same way. Whether the model is looking at a blank white corner of the canvas or a cluster of heavy, dark strokes right in the middle, the same weight parameters get applied across the board. For general image classification that's usually fine, but for analyzing children's drawings it's a real problem — the emotionally meaningful parts of a drawing are often localized to specific areas, and they can get drowned out by all the irrelevant background space around them.

Attention mechanisms were developed to address exactly this. The idea is to give the network a way to focus more on the parts of the image that actually matter and dial down the noise from the parts that don't.

One of the more interesting recent developments in this space is SimAM — the Simple Parameter-Free Attention Module. What makes it stand out compared to older attention approaches like SE-Net or CBAM is that it doesn't need any extra learned parameters to work. Those older modules require additional convolutional layers to figure out where to focus, which adds to the model's size and slows things down. SimAM takes a different approach: it computes attention weights dynamically by looking at how well a given neuron can be linearly separated from its neighbors, using an energy function to measure that. In plain terms, it identifies which parts of the feature map are locally distinctive and boosts those, without needing to learn anything new or add any extra weight to the model.

For drawing analysis this is particularly useful. A child who's feeling anxious might press hard in certain areas, leaving thick dark strokes, while a hesitant child might draw faint, tentative lines. SimAM helps the network pick up on these kinds of localized visual signals — things that a flat convolutional layer might otherwise smooth over — without making the model any heavier or slower to run.

2.3.3 Hierarchical and Multi-Model Fusion Architectures:

One of the challenges with emotion recognition is that emotions aren't clean, discrete categories — especially in children's drawings, where a single image might reflect a mix of feelings that don't map neatly onto one label. If you train a single neural network to output one emotion from a drawing, it tends to struggle with these ambiguous cases. The model gets pulled in multiple directions at once, and the result is often poor confidence scores and inconsistent predictions. The research literature suggests that the solution to this is some form of hierarchical or cascading architecture, where instead of asking one model to do everything, you break the problem down into stages.

The way a cascading setup typically works is that a primary model handles the first, broader discrimination in this case, separating generally positive emotional signals from negative ones. That's a relatively straightforward binary decision and a single model can do it reliably. But when the primary model isn't confident when the probability scores are close together or the output is genuinely ambiguous rather than just picking the higher of the two and moving on, the system passes the input to a second, more specialized model. That second model is trained specifically to distinguish between the subtler negative emotions that the first model couldn't resolve, like telling apart sadness, anger, and fear.

This design makes a lot of sense intuitively. Sadness and fear might both register as "negative" to a broad classifier, but they look quite different in a drawing and a model trained specifically on that distinction will do a much better job than one trying to handle everything at once. The cascading approach also helps with class imbalance, since each model in the chain only needs to handle a narrower slice of the emotional spectrum rather than competing across all categories simultaneously.

2.3.4 Large Language Models for Psychological Interpretation:

CNNs are genuinely good at picking up on visual patterns stroke textures, spatial arrangements, color distributions but that's basically where their usefulness ends. They can't read a parent's written notes, they don't understand context, and they certainly can't produce a paragraph of

warm, readable advice for a caregiver. For a system that's actually meant to be used by parents rather than researchers, that's a pretty significant gap.

The way modern clinical AI systems handle this is by pairing the vision model with a large language model. The CNN does what it's good at extracting numerical features from the image and then those features get handed off to an LLM, which does what it's good at: understanding context, combining multiple inputs, and generating coherent natural language.

The key to making this work well is prompt engineering. Rather than just dumping raw numbers into the language model and hoping for the best, the system structures the input carefully — packaging the visual feature values alongside the parent's observations and the child's own description into a formatted context block. This gives the model enough grounded, specific information to produce an output that's actually relevant to that particular child, rather than something generic.

One practical concern with LLM outputs is consistency. If the model is free to respond however it wants, the structure of the response can vary wildly from one call to the next, which makes it hard to parse and store reliably. Enforcing strict JSON output formatting solves this — by constraining the model to return a fixed set of fields, the application always gets back something it can work with directly, without needing fragile post-processing logic. It also helps keep the outputs grounded and factual, reducing the risk of the model generating confident-sounding but inaccurate clinical claims — which would be genuinely harmful in a child psychology context.

2.4 Related Existing Systems:

2.4.1 Existing System:

Nowadays, several specialized programs and applications have been developed to analyze children's drawings using artificial intelligence. These systems are specifically designed to assist in understanding the psychology of children by interpreting visual elements within their drawings. By relying on image processing, pattern recognition and machine learning models, such programs aim to provide early insights into a child's feelings, behavior, and mental well-being. Most of these systems are used by parents, educators, and psychologists as supportive tools rather than as definitive diagnostic solutions.

2.1.1 Problems and Weaknesses of Existing Systems:

Despite their benefits, existing systems suffer from several limitations.

1. That most applications require users to pay money to access full features or detailed analysis, which makes them unavailable for many parents.
- 2- The applications do not support the Arabic language. The few platforms that do support Arabic are usually limited to simple websites that only provide written explanations, rather than full intelligent applications.
3. Most existing systems analyze the child's drawings individually and do not allow for comparison or monitoring of the child's psychological development over time, which reduces the accuracy of the assessment and limits the ability to track psychological improvement or deterioration effectively.
4. These websites focus on analyzing only one child's drawing at a time and do not provide the ability to compare or analyze multiple children's drawings, which limits their ability to offer a comprehensive assessment or track the psychological state of several children

2.4.2 Comparison Table of Related Systems:

Functional Evaluation Parameter	System 1: Kido App (Mobile Platform)	System 2: Traditional Analytics Web Portals	System 3: Generic AI Chatbots (Unstructured)	Our Proposed System: Draw Mind AI
Multi-Profile Management	Limited to premium single accounts;	Non-existent; operates strictly on isolated	None; lacks structured relational	Fully Supported; multi-child demographic

	lacks collaborative cross-parent scaling.	session configurations	persistence layers.	tracking per user account.
Language Localization	Exclusively optimized for Western linguistic layouts; zero Arabic validation.	Hard-coded Arabic text options; no dynamic natural language engines.	Supports general Arabic text; lacks localized psychological understanding.	Native English dual interface support with contextual parsing.
Visual Feature Extraction	Basic image classification bounding loops; no fine-grained line textures.	Manual questionnaire input required from parents; no automated pixel tracking.	Completely blind to visual drawing structures unless fed with heavy prompts.	Automated processing of 14 visual indicators using customized CV pipelines.
Longitudinal Trend Tracking	Absent; treats uploaded artwork as independent, isolated interaction events.	No long-term historical database compilation over chronological timelines.	No long-term historical retention or cross-session memory tracking.	Supported via time-series persistent cloud database architectures.
Financial Access Barriers	Rigid paywalls; full report synthesis requires high premium subscriptions.	Ad-heavy monetization structures; limits advanced analysis dashboards.	Paywalled access limits token throughput for advanced image models.	Open educational access models tailored for universal caregiver scaling.
Core Technology Stack	Standard Convolutional Classifiers (Single-model topology).	Static HTML/JS form frameworks paired with rule-based scripts.	Generic Large Language Models lacking image preprocessing regularizes .	Decoupled CNN Cascading Models with SimAM Attention & Groq APIs.

Table 4: Comparison table of related systems

3. SYSTEM REQUIREMENTS ENGINEERING AND ANALYSIS

3.1 Introduction:

This chapter will cover the exploration process of the main elements for the success of developing and implementing our project.

It ensures that the developed system meets the needs of all stakeholders, is technically feasible, and works within the available cost and time. We will talk about the feasibility study, requirements elicitation techniques and the functional and non-functional requirements.

3.2 Feasibility Study:

In this section, we evaluate the applicability of the project from Technical, Operational, Economic, Schedule, and Legal perspectives to ensure that the proposed solution is viable and sustainable.

Feasibility Area	Assessment Criteria	Project Status / Strategy (DrawMind AI)	Feasibility Rating
Technical Viability	Availability of necessary frameworks, libraries, and APIs.	High. Utilizes established, open-source tech stacks: Python, FastAPI (Backend), TensorFlow/Keras (CNN models), OpenCV (Image processing), and Groq API (LLM integration).	Highly Feasible
Hardware Constraints	Processing power required for training and inference.	High. Training was completed off-site (Google Colab). Inference utilizes lightweight, optimized models (MobileNetV2, EfficientNetV2-S) capable of running efficiently on standard server architecture without requiring high-end GPUs.	Highly Feasible
Development Expertise	Team proficiency with required languages and architectures.	Moderate-to-High. Requires knowledge of cascading machine learning architectures, API integration, and full-stack development, all of which are demonstrated in the current prototype.	Feasible
Development Costs	Initial costs for software, tools, and development time.	Low. Relies entirely on open-source libraries and frameworks. Primary costs are limited to developer time and initial low-tier hosting environments.	Highly Feasible

Operational Costs	Ongoing expenses (Hosting, API usage, Database storage).	Moderate. Groq API introduces variable costs based on usage. Hosting (e.g., AWS, Heroku) and database storage (time-series tracking) will scale linearly with user growth.	Feasible
Market Viability	Value proposition compared to existing market solutions.	High. Offers automated visual feature extraction and dual-language (Arabic/English) support, significantly outperforming competitors like Kido App or generic AI chatbots.	Highly Feasible

Table 5: Feasibility Study

3.2.1 Technical Feasibility:

The technical feasibility of this project is established by the availability of mature, open-source technologies and the high-performance frameworks required to handle complex image analysis. Our proposed system is designed as a first mobile application to provide an intuitive drawing experience for children through a touch-responsive interface. This approach ensures that data collection, whether through direct digital drawing or photo uploads—is practical and user-friendly.

From an AI perspective, the system’s core functionality is built on **Convolutional Neural Networks (CNNs)**. Instead of training a model from scratch, we will utilize **Transfer Learning** using architectures like **MobileNet** or **ResNet**, which are specifically optimized for mobile environments and high-accuracy feature extraction. These models will allow the application to objectively analyze visual markers—such as color density, shape repetition, and spatial layout—which traditionally require subjective manual interpretation.

The backend will be developed using **Python** (via **FastAPI** or **Flask**) to manage the heavy computational logic of the AI models. Since we are dealing with sensitive psychological data, we have prioritized security; the system will use encrypted communication between the mobile app and the server, with data stored in a secure **MongoDB** or **Firebase** environment.

All necessary hardware, including tablets for the children and development workstations for the team, is already accessible and within our current resources. This ensures that the technical implementation can be completed successfully within the planned project timeline.

- Key Technologies and Tools:

Programming & Logic: Python (FastAPI/Flask) for scalable backend services.

Artificial Intelligence: TensorFlow and PyTorch for implementing and fine-tuning the CNN models.

Data Storage: MongoDB Atlas or Firebase for secure, real-time data management.

Mobile Frontend: Flutter or React Native to ensure smooth, cross-platform user experience.

Computer Vision: OpenCV and Pillow for essential image preprocessing, such as noise reduction and resizing.

3.2.2 Operational Feasibility:

The system is operationally feasible as it is designed to be intuitive, child-friendly, and easy to use.

Children can interact with the application by drawing freely on a digital canvas or uploading existing drawings without requiring technical knowledge.

Parents, teachers, or caregivers can view the system's output in a simple and understandable format, which provides supportive insights into the child's emotional or psychological state. The application does not require special training to operate, making it suitable for use in homes, schools, and educational centers.

The system integrates smoothly into daily routines without disrupting normal activities. By automating the analysis process, the application reduces the need for continuous manual observation while supporting early awareness of emotional patterns in children.

This operational approach enhances usability while maintaining ethical and educational considerations.

Operational Parameter	Assessment Focus	DrawMind AI Implementation & Outcome
User Adoption & Accessibility	How easily can the target audience use the system?	Strong. Designed with an accessible frontend interface. Eliminates the need for parents to manually answer complex psychological questionnaires by automating feature extraction from a simple photo upload.
Workflow Integration	Does it fit into the existing routines of users (e.g., schools/clinics)?	Strong. Provides instant, downloadable psychological reports that can be filed in a student's record or shared directly with a professional, seamlessly integrating into existing diagnostic workflows.
System Reliability	How does the system handle errors, bad data, or ambiguous inputs?	Strong. Employs a cascading CNN architecture. If the primary model is "Not Sure" or detects complex emotion blending (e.g., Sad + Fear), the system gracefully handles the ambiguity and feeds this context to the Groq API for a nuanced, non-binary explanation.
Language & Cultural Competence	Can the system serve the intended local demographics?	Strong. Native dual-language (Arabic and English) support ensures accessibility for regional users, a feature entirely missing from current Western-centric competitors.

Maintenance & Scalability	Can the system be maintained and updated easily over time?	Moderate. The decoupled architecture (FastAPI backend separate from the Groq API and Frontend) allows individual components to be updated or swapped (e.g., changing the LLM provider) without breaking the entire system.
Ethical & Clinical Boundaries	Does the system overstep clinical boundaries or misdiagnose?	Strong (Mitigated). The prompt engineering strictly instructs the LLM to <i>never</i> make clinical diagnoses. It provides warm, supportive guidance and suggestions, acting as a triage tool rather than a replacement for human professionals.

Table 6: Operational Feasibility

3.2.3 Economic Feasibility:

Year	Development Cost (JD)	Operating Cost (JD)	Total Cost (JD)
Year 0	200	350	550
Year 1	0.00	400	500
Year 2	0.00	450	450
Year 3	0.00	500	500
Year 4	0.00	550	550
Year 5	0.00	600	600

Table 7:Economic Feasibility

3.2.4 Schedule Feasibility:

Tas k #	Task Name	Descriptio n	Start Time	End Time	Duratio n	Dependencie s
T #1	Gather Project Require ments	Collect and define project goals, ...	1/11/2025	8/11/2025	7 Days	None
T #2	Project Planning	Plan project timeline and tasks	9/11/2025	12/11/2025	3 Days	T #1
T #3	Design System Architect ure	Plan app and backend structure	13/11/2025	19/11/2025	6 Days	T #1, T #2
T #4	Design AI Workflow	Define AI process and data flow	20/11/2025	26/11/2025	5 Days	T #1, T #2

T #5	Develop Backend & Database	Implementing server and database	27/11/2025	3/12/2025	7 Days	T #3
T #6	Gathering Data	Collect and label drawings	4/12/2025	13/12/2024	9 Days	T #1, T #4
T #7	Train Deep Learning Model	Train AI mode	14/12/2025	23/12/2025	9 Days	T #6
T #8	Integrate AI with App	Connect model with app	24/12/2025	28/12/2025	5 Days	T #5, T #7
T #9	Design User Interface	UI design for children	29/12/2025	02/1/2026	5 Days	T #1
T #10	System Testing	ent Integrate all components (database...)	3/01/2026	9/01/2026	7 Days	T #8, T #9
T #11	Deployment	Deploy app for use	10/1/2026	14/1/2026	5 Days	T #10

- **Description** for each task:

T1 Gather Project Requirements Collect and define the overall project goals, user needs, and functional requirements for the application. This involves interviewing stakeholders, understanding objectives, and documenting specifications.

T2 Project Planning Develop a detailed project plan including timeline, task assignments, milestones, and risk management strategies.

T3 Design System Architecture Plan the overall system structure including mobile app, backend server, database schema, and component interactions.

T4 Design AI Workflow Create the workflow for drawing collection, image preprocessing, AI model training, and generating predictions.

T5 Develop Backend & Database Implement backend server and database to store children's drawings, user data, and AI results securely.

T6 Data Collection Collect and prepare children's drawings, label them correctly, and organize the dataset for AI training and testing.

T7 Train Deep Learning Model Train and evaluate the AI model for classifying psychological states. Includes selecting CNN architectures, tuning parameters, and validating performance.

T8 Integrate AI with App Connect the trained AI model with the mobile app to ensure real-time predictions. Includes API development and integration testing.

T9 Design User Interface Design a child-friendly interface for drawing input and result visualization. Includes wireframes, layout, and interaction flow.

T10 System Testing Conduct functional, usability, and performance testing of the app. Identify and fix bugs, ensure AI accuracy, and smooth app performance.

T11 Deployment Deploy the app to production/demo environment and provide instructions for users. Ensure system stability and readiness for demonstration or real use

3.2.5 Legal Feasibility:

Ensuring full compliance with data protection and privacy regulations is a fundamental requirement for the legal and safe deployment of the Child Psychological State Recognition application. In the Jordanian context, the **Personal Data Protection Law (PDPL)** serves as the primary legal framework that governs the collection, storage, and processing of personal information. Since our system handles sensitive digital content—specifically children’s drawings and psychological indicators—it is imperative to safeguard this data against unauthorized access, misuse, or potential breaches.

To align with PDPL standards and prepare the system for official licensing by the Ministry of Higher Education, the following measures are integrated into the system's architecture:

Advanced Encryption Protocols: Implementing robust data encryption for information both in transit and at rest to maintain high levels of confidentiality.

Controlled Access & Monitoring: Utilizing secure database management systems with restricted access controls and continuous monitoring to prevent data leakage.

Informed Consent Procedures: Establishing a clear process for obtaining guardian consent, which provides a transparent explanation of how drawings will be analyzed and how the results will be utilized.

Data Subject Rights Management: Providing guardians with the absolute right to access, rectify, or permanently delete their child’s information from the system, ensuring full control over their personal data.

3.3 The Requirements Gathering Techniques:

The requirements of the Draw Mind AI system were identified using a combination of analytical and design-driven techniques. An extensive review of existing research and academic literature related to children's emotional analysis through drawings was conducted to understand commonly used visual features, emotion categories, and system functionalities. This review helped establish a theoretical foundation for the system and ensured alignment with scientifically recognized approaches.

In addition, existing emotional analysis and mental health support applications were examined to identify their strengths, limitations, and missing functionalities. This analysis contributed to defining features that address usability gaps and improve interpretability for non-technical users. Furthermore, realistic usage scenarios were considered to model how professionals such as parents, teachers or psychologists would interact with the system, including uploading drawings, viewing analysis results, and tracking emotional progress over time.

3.4 Targeted Customers:

The Draw Mind AI system is primarily designed for individuals who interact closely with children in their daily environments, including parents, caregivers, teachers, and guardians. These users often observe children's drawings but may lack the expertise or tools needed to interpret the emotional meaning behind them. The system aims to support these users by providing accessible, easy-to-understand emotional insights derived from children's drawings, helping them better recognize potential emotional states and patterns over time.

The application is intentionally designed to be simple and intuitive, requiring no technical or psychological background. Visual explanations, clear feedback, and progress tracking features are provided to assist users in gaining preliminary emotional awareness rather than clinical assessment. While professionals such as psychologists or counselors may find the system useful as a supportive reference tool, it is not intended for diagnostic purposes and does not replace professional psychological evaluation. The system is also not designed for direct use by children.

Table below presents the key stakeholders of the Draw Mind AI system and summarizes their roles, interactions, and interests.

Stakeholder	Role	Interaction with System	Interest / Benefit
Parents / Guardians/ Caregivers	Primary user	Upload drawings, view emotional analysis and progress	Better understanding of child's emotional state. Support daily emotional observation.
Teachers	Primary user	Analyze drawings and monitor emotional patterns.	Early awareness of emotional concerns.
Children	Indirect stakeholder	Provide drawings only.	Emotional expression through drawing.
Mental Health Professionals	Secondary stakeholder	Review outputs if needed.	Supportive reference (non-diagnostic)

Table 8:Stakeholders of the System

3.5 Functional Requirements:

Functional Requirements describe what the system should do.

They define the specific functions, services, and behaviors that the system must provide to its users. These requirements specify how the system responds to particular inputs, how it processes data, and how it interacts with users and other system components.

FR ID	Functional Requirement	Description	Stakeholder
FR-1	User Registration	The system shall allow users to create a new account.	Parents / Guardians / Caregivers
FR-2	User Login	The system shall allow registered users to log in using valid credentials.	Parents / Guardians / Caregivers
FR-3	User Logout	The system shall allow users to log out of the system securely.	Parents / Guardians / Caregivers
FR-4	Password Recovery	The system shall allow users to recover their password if forgotten.	Parents / Guardians / Caregivers
FR-5	Change Password	The system shall allow users to change their password.	Parents / Guardians / Caregivers
FR-6	Change Language	The system shall allow users to change the system language.	All Users
FR-7	Add Child Profile	The system shall allow parents to add a child profile.	Parents / Guardians / Caregivers
FR-8	Update Child Information	The system shall allow parents to update child information.	Parents / Guardians / Caregivers
FR-9	Drawing Tool	The system shall allow children to draw directly on the application.	Children
FR-10	Upload Drawing Image	The system shall allow users to upload images of children's drawings.	Parents / Guardians / Caregivers
FR-11	Capture Drawing Image	The system shall allow users to capture images using the device camera.	Parents / Guardians / Caregivers
FR-12	Text Input	The system shall allow parents to enter behavioral observations about the child.	Parents / Guardians / Caregivers
FR-13	Analyze Drawings	The system shall analyze children's drawings using artificial intelligence techniques.	System
FR-14	Analyze Text Input	The system shall analyze the entered text related to child behavior.	System
FR-15	Display Analysis Results	The system shall display the analysis results in a clear and understandable format.	Parents / Guardians
FR-16	Request Help	The system shall allow parents to request help or consultation.	Parents / Guardians

FR-17	View Previous Results	The system shall allow users to view previously analyzed drawings and their results for each child.	Parents / Guardians / Caregivers
FR-18	Emotional Trends Visualization	The system shall display a chart showing the child's emotional changes over time based on previous analysis results.	Parents / Guardians / Caregivers
FR-19	View Child Profile	The system shall allow users to view the child's profile, including personal information and analysis history.	Parents / Guardians / Caregivers
FR-20	Intelligent Assistant (Chatbot)	The system shall allow users to interact with an intelligent assistant (chatbot) to receive psychological insights and guidance based on the child's emotional analysis.	Parents / Guardians / Caregivers
FR-21	Store Analysis Results	The system shall store analysis results and link them to the corresponding child profile for future access.	System
FR-22	View Doctor Contact Information	The system shall provide users with contact information for specialized doctors.	Parents / Guardians / Caregivers

Table 9: Functional Requirements

3.6 Non-Functional Requirements:

NFR ID	Non-Functional Requirements	Description
NFR-1	Performance	The system processes and analyzes drawings efficiently with fast response time and optimized resource usage (CPU, memory, and battery).
NFR-2	Scalability	The system supports future growth by handling an increasing number of users, children profiles, and analysis data without performance degradation.
NFR-3	Security & Privacy	User data and children's drawings are securely stored and protected to prevent unauthorized access, ensuring privacy compliance.
NFR-4	Reliability	The system operates consistently with minimal failures and provides stable performance during usage.
NFR-5	Availability	The application remains accessible most of the time,

		with proper error handling and user-friendly messages in case of system or network issues.
NFR-6	Maintainability	The system is built using a modular structure with clear documentation to allow easy updates, debugging, and future enhancements.
NFR-7	Accuracy	The AI model provides reliable and reasonable predictions based on trained datasets of children's drawings.
NFR-8	Usability	The interface is simple, intuitive, and designed for non-technical users such as parents, ensuring easy navigation and interaction.
NFR-9	Ethics & Responsible AI	The system provides supportive emotional indicators rather than medical diagnoses, and includes clear disclaimers recommending professional consultation when needed.
NFR-10	Transparency	The system clearly explains that results are AI-based interpretations and not final judgments, ensuring user trust.
NFR-11	Personalization	The system supports multiple child profiles, each with separate history and analysis results.

Table 10: Non-Functional Requirements

3.7 Dataset Construction, Challenges, and Merging Strategy:

One of the most consequential engineering decisions in developing this assistive, preliminary analytical tool was the strategy used to construct a domain-appropriate training dataset. No single publicly available dataset provided sufficient volume, class diversity, and domain specificity for a four-class children's emotional drawing classifier. The dataset construction therefore involved a deliberate, multi-source assembly process governed by the distributional requirements of the hierarchical model architecture.

3.7.1 The Core Data Problem: Scarcity, Imbalance, and Domain Mismatch:

The dataset construction challenge in this project extends beyond simple data scarcity. Children's drawing datasets with emotional labels are rare, and those that exist are heavily skewed toward positively valenced emotions, particularly happiness. Negative emotions such as anger and fear

are represented at a fraction of the volume of positive emotions in all publicly available collections. Furthermore, the visual domain of children's emotional drawings presents challenges that are absent in standard image classification benchmarks. The emotional signal in these drawings is carried not by photorealistic cues but by abstract visual conventions—the size of figures, the presence or absence of color, the heaviness of strokes, and the spatial organization of elements.

3.7.2 Why a Single Dataset Was Not Sufficient:

The decision to train two specialized binary models rather than a single four-class model was driven directly by the dataset landscape. Training a single four-class model under the conditions of severe class imbalance would produce a classifier biased toward the majority classes, with systematically poor recall on anger and fear. The architectural solution adopted in this project was to separate the classification problem along the valence dimension. A dedicated binary model was assigned to the positive/negative valence pair of happiness and sadness, where sufficient training data exists in a single unified dataset. A second binary model was assigned exclusively to the anger/fear distinction, which required data to be assembled from multiple sources.

3.7.3 Dataset 1 — KIDO Children's Drawing Dataset (Happy/Sad Model):

The first binary model was trained on the KIDO Children's Drawing Dataset, accessed via the Kaggle platform. The KIDO dataset is a multi-dimensional collection organized by several annotation axes. For this project, only the emotional annotation axis was used; the gender and text portions of the dataset were ignored entirely. Within the emotion category, only the Happiness and Sadness classes were extracted.

The dataset contains a total of 10,860 images distributed equally between the two classes, with 5,430 images per class. This absolute balance is a significant advantage for binary classification and eliminated the need for class weighting at the dataset level. The dataset was split using a stratified random sampling strategy with a fixed random seed (42) to ensure reproducibility. The 80/10/10 split produced 8,688 training images, 1,086 validation images, and 1,086 test images. The training set was subsequently augmented online during training.

Split	Happiness	Sadness	Total
Training (80%)	4,344	4,344	8,688
Validation (10%)	543	543	1,086
Test (10%)	543	543	1,086
TOTAL	5,430	5,430	10,860

Table 11: Distribution of KIDO dataset images across training, validation, and test splits for the Happiness/Sadness model.

3.7.4 Dataset 2 & 3— Combined Angry/Fear Dataset :

The construction of the training dataset for the anger/fear binary model was the more technically demanding aspect of the data engineering process. Two complementary sources were identified and combined.

The first source was the Children's Drawings dataset contributed by Vishmi Perera (Kaggle), containing approximately 544 images. This dataset was not originally labeled for emotional content; instead, images that could be associated with the angry and fear emotional categories through their visual content were retained and manually assigned the appropriate label. The second source was the Emotion in Drawings dataset hosted on Roboflow Universe. For this model, only the angry and fear classes were extracted, providing 240 angry images and 249 fear images across all splits.

Split	Angry	Fear	Happy*	Sad*	Total
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Train	174	172	169*	139*	654
Valid	41	55	49*	42*	187
Test	25	22	19*	28*	94
Extracted	240	249	—	—	489

Table 12: Full distribution of the Roboflow dataset. Classes marked with an asterisk (*) were excluded from the anger/fear model training.

The two sources were merged at the file path level, producing an initial combined pool of 1,033 images. Prior to processing, a deduplication scan was performed using MD5 cryptographic hashing. This process identified and removed 123 exact duplicate images that appeared across both source datasets, reducing the clean dataset to 910 images.

Source	Angry	Fear	Subtotal	Duplicates Removed	Clean Images
Kaggle (Vishmi Perera)	267	277	544	—	—
Roboflow (Emotion in Drawings)	240	249	489	—	—
Combined (pre-dedup)	507	526	1,033	123	910
Final Clean Dataset	451	459	910	—	910

Table 13: Dataset construction summary for the Angry/Fear model after MD5-based deduplication.

Table 14: Datasets.

Dataset ID	Dataset Name	Classes Used
Dataset 1	Kido Children Drawing Dataset	Happy, Sad
Dataset 2	Children Drawings Dataset	Angry, Fear
Dataset 3	Emotion in Drawings Dataset	Angry, Fear

3.7.5 The Inherent Complexity of the Visual Domain: Why These Datasets Are Genuinely Difficult:

The dataset construction challenge extends beyond logistical sourcing and class imbalance; it is fundamentally rooted in the psychological and visual ambiguity of the data itself. This project does not tackle standard object recognition (e.g., identifying a cat or a car), where universal geometric rules apply. Instead, it attempts to quantify abstract psychological expression. This presents four severe computational and domain-specific challenges:

1. The "Zero-Context" Limitation of Pure Vision:

Clinical psychological analysis rarely relies on a drawing in isolation. A human therapist utilizes essential contextual information, including the child's background, baseline behavior, and verbal explanations of the artwork. The core vision models, however, are forced to perform a "zero-context" analysis during the extraction phase, evaluating patterns exclusively from pixel distributions. This inherent limitation of pure computer vision is precisely what necessitated the integration of the Grok LLM and Explainable AI

(XAI) modules—discussed in subsequent sections—into the final application. These modules act to synthesize the network's raw visual predictions with the contextual and behavioral data inputted by the user, ensuring the system functions as a robust, contextualized assistive tool rather than a standalone diagnostic system.

2. Severe Visual and Geometric Ambiguity (The Visual Evidence):

Standard image classification benchmarks rely on structural predictability. For example, facial emotion recognition algorithms depend on fixed distances between the eyes and the mouth. Children's drawings, however, are highly abstract and lack standardized spatial features.

Figure 1: A collage showing the provided KIDO dataset images — e.g., the pink bus, the red cartoon character, the green-topped square and the floating scribbles, labeled as "Happy".

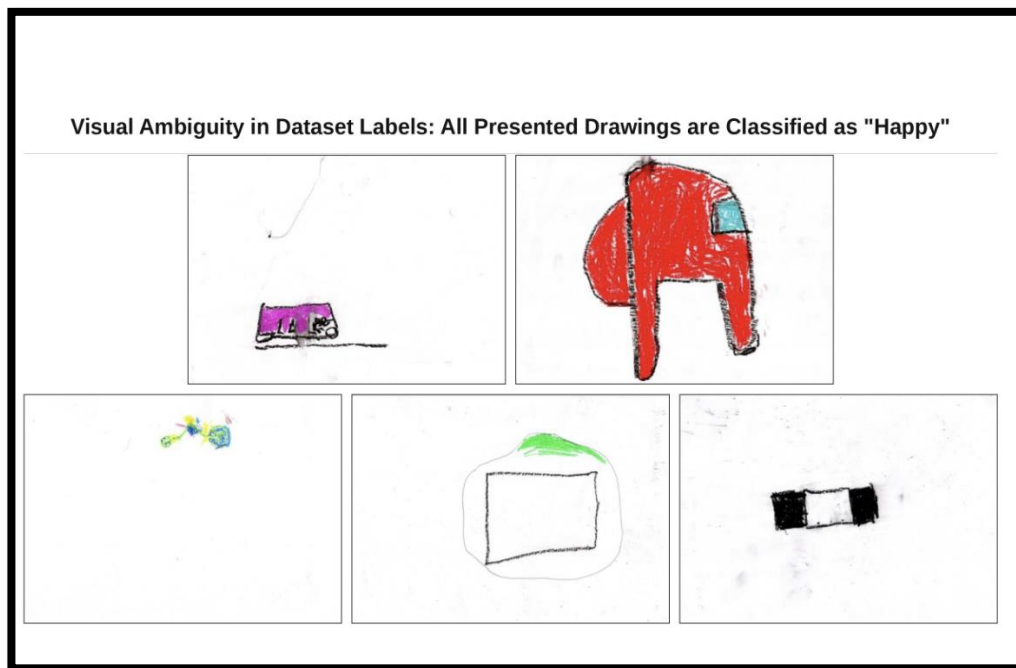
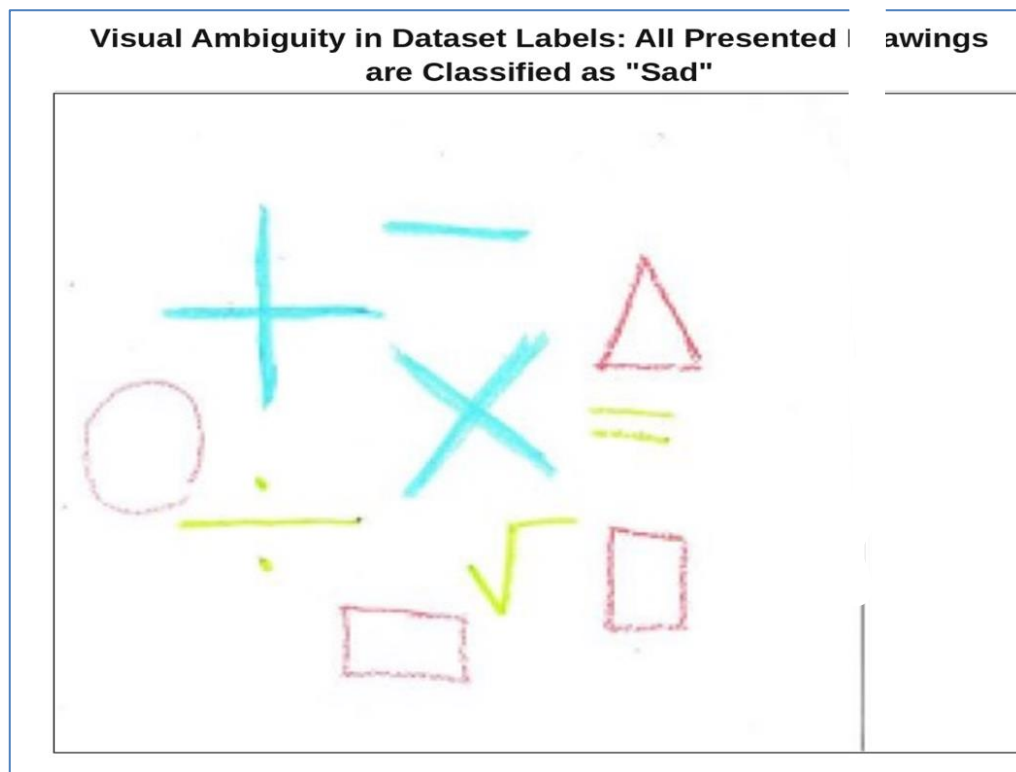


Figure 2: A counter-example from the KIDO dataset showing disconnected mathematical symbols and abstract shapes, ground-truth labeled as "Sad", highlighting the severe visual ambiguity between classes.



As explicitly demonstrated in the KIDO dataset samples (Figure 1) and (Figure 2), elements frequently float, sizes are disproportionate, and generic or abstract shapes lack explicit markers of emotion.

A neural network is fed a drawing of an isolated pink vehicle, a red cartoon character, or random floating geometric shapes, all of which represent the "Happiness" class. Conversely, a visually similar abstract shape—such as a heavily shaded black rectangle or disconnected mathematical symbols—represents the "Sadness" class. The human brain itself struggles to find a definitive emotional meaning in these disconnected drawings without the child's accompanying narrative, making it exceptionally difficult for a CNN to learn distinct feature maps.

3. The Subjectivity of "Ground Truth" Labels:

In standard AI datasets, an image label is objective and undeniable. In the domain of psychological drawings, labels are inherently subjective. Training a deterministic mathematical model on data where the "ground truth" is abstract—where a child's

drawing of a simple green-and-black square is definitively labeled as "Happy" by the dataset creators—requires an exceptionally robust preprocessing strategy to find the hidden structural patterns beneath the seemingly random pixel arrangements.

4. The Psychological Overlap of High-Arousal Emotions:

Distinguishing specifically between Anger and Fear presents a severe psychological and computational challenge. Both are high-arousal, negative-valence emotions that frequently manifest through overlapping abstract visual conventions—such as chaotic scribbles, heavy stroke pressure, and a complete lack of spatial organization. Sourcing reliable data for these classes required navigating multiple datasets and performing rigorous MD5 deduplication. Furthermore, standard pretrained CNN features are not natively optimized to untangle these nuanced, overlapping visual cues from abstract strokes, underscoring the absolute necessity of the custom Master Filter preprocessing pipeline to isolate the true structural signal from the noise.

3.7.6 Comparative Overview of Both Datasets:

In conclusion, the dual-dataset strategy successfully mitigated the fundamental issue of data scarcity and severe class imbalance. The first dataset (KIDO) provided a massive, perfectly balanced foundation of 10,860 images to train the foundational positive/negative valence extraction model. In contrast, the engineered Angry/Fear dataset utilized targeted aggregation and cryptographic deduplication to yield a highly specialized, balanced subset of 910 clean images. By isolating the data domains according to the target classification task, the architectural design ensures that neither model is compromised by the distributional flaws of the global dataset landscape.

3.8 Master Preprocessing Filter:

A uniform preprocessing pipeline, referred to throughout this project as the Master Filter, was applied to every image entering either base model, including both training and inference paths. The design of this pipeline was motivated by the specific visual characteristics of children's drawings, which differ substantially from the natural image domain on which pretrained models such as MobileNetV2 are originally trained.

Children's drawings typically consist of sparse ink or pencil marks on a white or near-white background. The emotional signal is concentrated in the stroke structure and spatial organization of drawn elements, not in color gradients or texture. Raw images of such drawings often contain noise introduced by the capture device (camera shake, lighting variation, paper texture), which can obscure the stroke-level features that the model must learn to associate with emotional categories. The Master Filter addresses these characteristics through a three-stage processing chain.

3.8.1 Stage 1: Bilateral Filtering for Noise Suppression:

The first stage applies a bilateral filter, parameterized empirically with a spatial neighborhood diameter of five pixels and variance thresholds (color space sigma of 50, spatial sigma of 50) tuned specifically to the stroke frequencies found in the dataset. The bilateral filter is a non-linear, edge-preserving smoothing operation that reduces high-frequency noise in flat regions, such as the

paper background, while maintaining the sharpness of edge transitions, such as drawn strokes. Unlike standard Gaussian blurring, which degrades edge sharpness uniformly, the bilateral filter weights the contribution of neighboring pixels by both spatial proximity and intensity similarity. This mathematical property makes it exceptionally well-suited for children's drawings, where the stroke-background boundary is the primary carrier of semantic and psychological information and must be strictly preserved during denoising.

3.8.2 Stage 2: Sharpening via Custom Convolution Kernel:

Following Sharpening via Custom Convolution Kernel Following noise reduction, the second stage applies a custom sharpening kernel through a two-dimensional convolution operation. The kernel utilized is a modified Laplacian enhancement filter, defined as:

$$K = \begin{bmatrix} 0 & -0.5 & 0 \\ -0.5 & 3.0 & -0.5 \\ 0 & -0.5 & 0 \end{bmatrix}$$

This kernel amplifies the central pixel intensity while subtracting a weighted contribution from its four orthogonal neighbors. This produces a net sharpening effect that significantly enhances the contrast between the stroke edges and the background regions. The central coefficient of 3.0 was selected to produce a moderate sharpening effect that reinforces stroke definition without introducing ringing artifacts in regions with closely spaced strokes. Following the convolution, all pixel values are clipped to the valid 8-bit intensity range of $[0, 255]$ to prevent numerical overflow. The mathematical transformation of these pixel intensities is explicitly validated through comparative pixel-level heatmap visualizations. The raw numerical vectors (see Figure 3.3)

display significant background variance and noise, which is successfully normalized and geometrically sharpened in the post-processing vectors (see Figure 3.4).

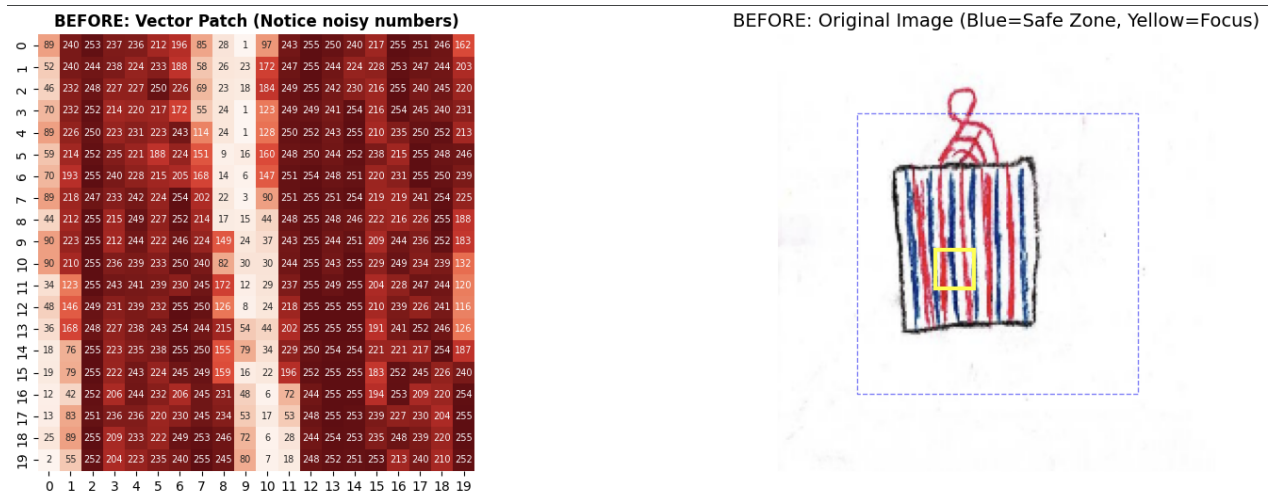


Figure 3.3 Pixel-level heatmap visualization of a raw, unprocessed drawing patch, exhibiting significant background noise and numerical variance

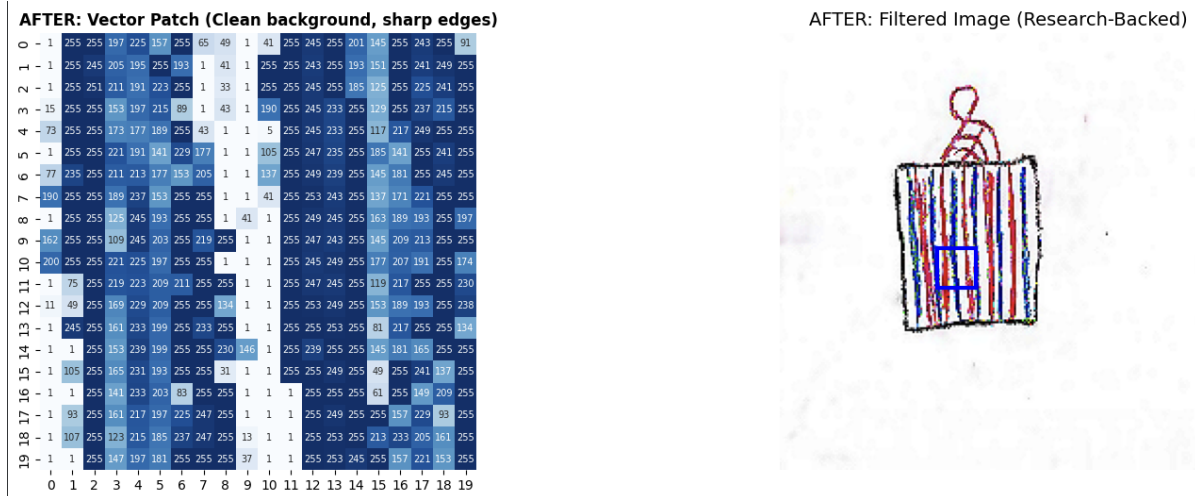


Figure 3.4 Pixel-level heatmap visualization of the drawing patch after applying the Master filter demonstrating normalized background intensity (255) and sharpened stroke structures.

3.8.3 Stage 3: Tensor MobileNetV2 Normalization:

The third and final stage applies domain-specific tensor normalization to rescale the pixel intensities from the 8-bit $[0, 255]$ integer range to a $[-1.0, 1.0]$ floating-point range. This transformation is executed using the exact mathematical mapping required by the MobileNetV2 architecture: $f(x) = (x / 127.5) - 1.0$. This strict normalization is mandatory to ensure mathematical compatibility with the pretrained ImageNet weights utilized in the base models, which were originally optimized for this specific input distribution. Executing this normalization as the terminal step—after bilateral filtering and spatial sharpening—guarantees that the custom morphological enhancements do not violate the expected statistical distribution of the pretrained feature extractor.

To maintain strict data integrity, this complete three-stage pipeline was applied deterministically to all validation and test partitions. During the training phase, stochastic data augmentation operations (including horizontal flipping, affine rotations, and photometric brightness/contrast jittering) were applied strictly to the raw 8-bit image matrices prior to the Master Filter. This specific execution order ensures that the filter processes the spatial distortions natively, preventing the final normalized bounds from being corrupted by post-normalization photometric shifts. The cumulative effect of this three-stage Master Filter pipeline significantly alters the visual and structural properties of the input data across all emotional categories. This consistent enhancement is explicitly validated through before-and-after visual comparisons of the four target emotions:

Happiness (see Figure 5.3), Sadness (see Figure 3.6), Anger (see Figure 3.7), and Fear (see Figure 3.8).

As demonstrated in these figures, the raw drawings initially exhibit substantial background noise and faint strokes, whereas the final processed images present isolated, high-contrast structural features optimal for convolutional feature extraction.

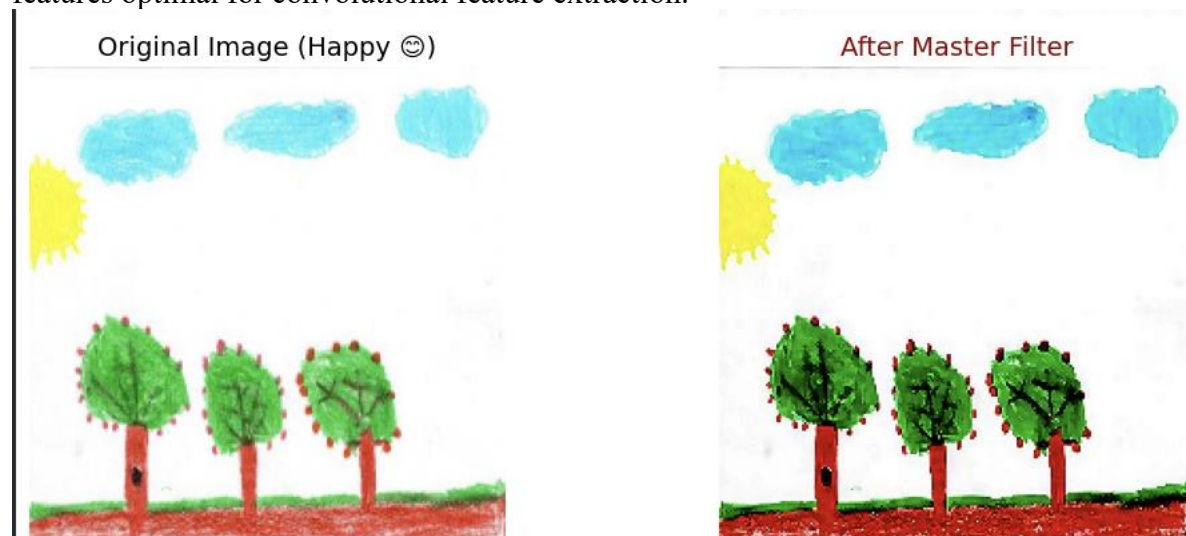


Figure 5.3 Master Filter before-and-after visual comparison demonstrating noise suppression and edge enhancement for a drawing from the Happiness category.



Figure 3.6 Master Filter before-and-after visual comparison demonstrating noise suppression and edge enhancement for a drawing from the Sadness category.

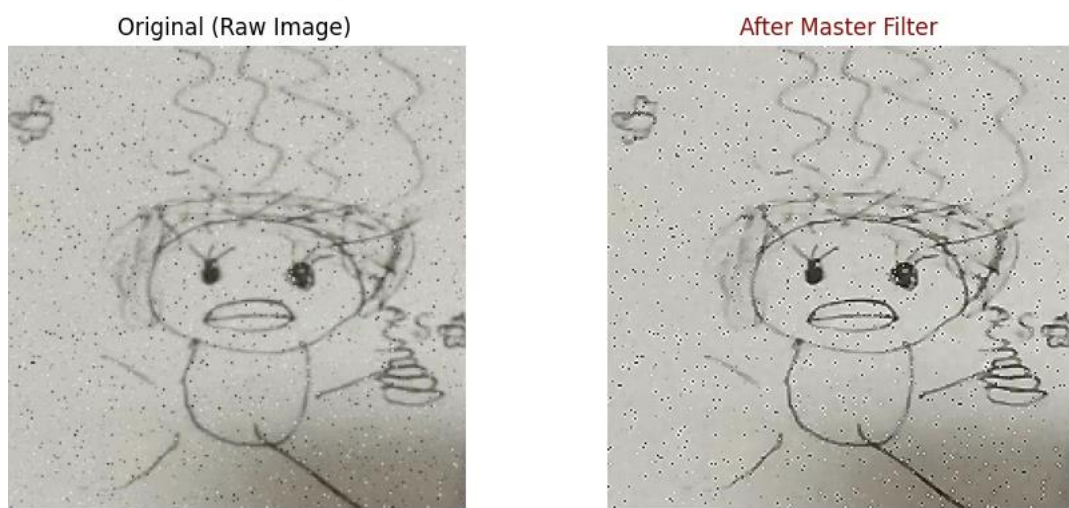


Figure 3.7 Master Filter before-and-after visual comparison demonstrating noise suppression and edge enhancement for a drawing from the Anger category.

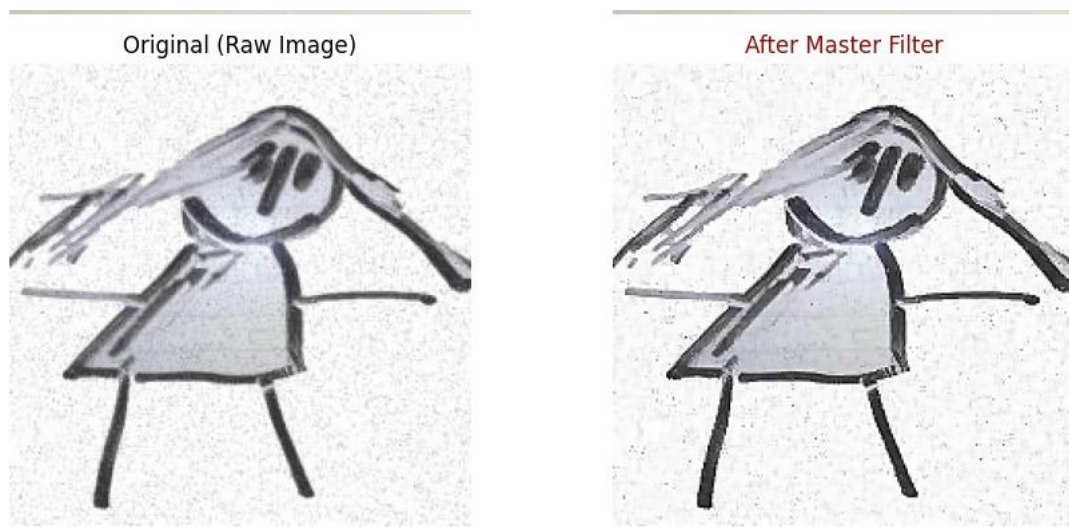


Figure 3.8 Master Filter before-and-after visual comparison demonstrating noise suppression and edge enhancement for a drawing from the Fear category.

3.11: Testing Strategy:

Before the platform goes live, a proper testing strategy needs to be in place to make sure everything actually works the backend processes, the API calls, the database connections, all of it. Given how much the system relies on asynchronous operations and cross-service communication, testing was broken down into three main phases: Unit Testing, Integration Testing, and Data Validation Testing.

3.11.1 Unit Testing Methodology:

Unit testing is basically about checking each small piece of the backend on its own, before anything talks to anything else. Within the FastAPI framework, these tests are designed to run without needing live databases or external services. The main things being tested here are: Endpoint Routing Integrity: Each HTTP route is checked individually including the authentication endpoints and the paths users take when submitting data, to make sure requests are handled and directed correctly.

Isolated Database Operations: The ORM models are tested to confirm they can correctly parse and handle data types before anything actually gets written to the cloud database. This catches structural issues early before they cause real transaction errors.

Feature Extraction Core Functions: The mathematical functions responsible for analyzing visual elements in children's drawings are tested to confirm they produce consistent, repeatable outputs every time they run.

3.11.2 Integration Testing Methodology

Once the individual parts are verified, integration testing checks how well they communicate with each other. The goal is to catch any synchronization issues or data mismatches that only appear when subsystems are actually connected. The main pipelines tested are:

Client-to-Server File Uploads: This tests what happens when a drawing file travels from the mobile app all the way to the FastAPI server specifically checking that multipart form data is correctly serialized and received without corruption or loss.

Vision-Language Model API Pipeline: This verifies the full chain of assembling a prompt combining child demographic data, parental notes, and visual feature arrays and then checks that the response coming back from the Gemini API is structured as expected.

Cloud Infrastructure Sync: This checks the connection between the local server and the remote PostgreSQL database, making sure there are no transaction deadlocks or schema conflicts, especially under heavier query loads.

3.11.3 Data Validation and Exception Handling Strategy

On top of unit and integration testing, a data validation layer is built directly into the input-handling models. This acts as a first line of defense at the gateway level, screening incoming requests before they reach any core processing logic. The key areas covered are:

MIME-Type and Extension Verification: Any file uploaded to the system is checked against a strict whitelist of accepted formats; only valid image types like JPEG and PNG are allowed through. Anything corrupt or unrecognized gets rejected immediately.

Linguistic Bound Restrictions: Text fields submitted through the parental input module are validated to catch things like empty strings or inputs that exceed the character limit, before anything gets passed on to the language model.

Graceful Degradation and Fallbacks: Try-catch blocks are wrapped around all third-party dependencies, so if the database times out or the Gemini API returns an unexpected error, the application handles it cleanly rather than crashing outright

4. SYSTEM DESIGN

4.1 Introduction:

This chapter outlines the system's structural and behavioral components through a series of standardized engineering models, including the Context Diagram, Data Flow Diagrams (DFD), and the Entity Relationship Diagram (ERD) to define data movement and storage logic.

Additionally, it details user interactions and system processes using UML Use Case and Sequence Diagrams, alongside a presentation of the Graphical User Interface (GUI) through low-fidelity prototypes. By establishing these design specifications, the chapter ensures that the AI-driven emotional analysis is integrated into a coherent, scalable, and user-centric technical framework.

4.2 Context Diagram:

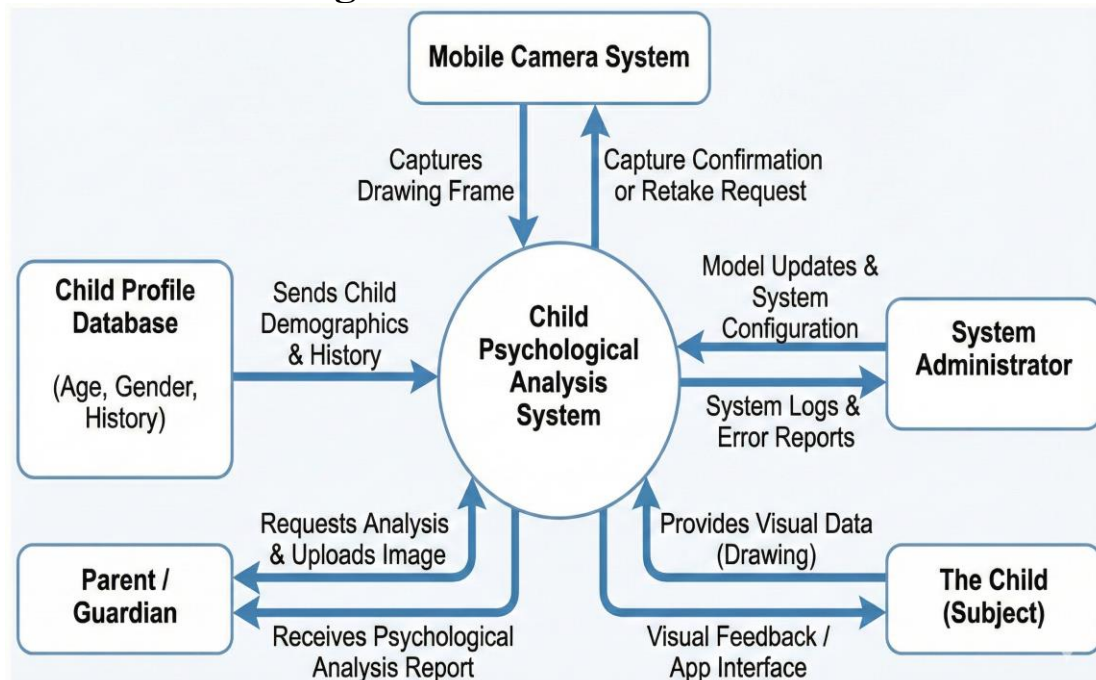


Figure 9: Context Diagram

4.3 Data Flow Diagram (DFD):

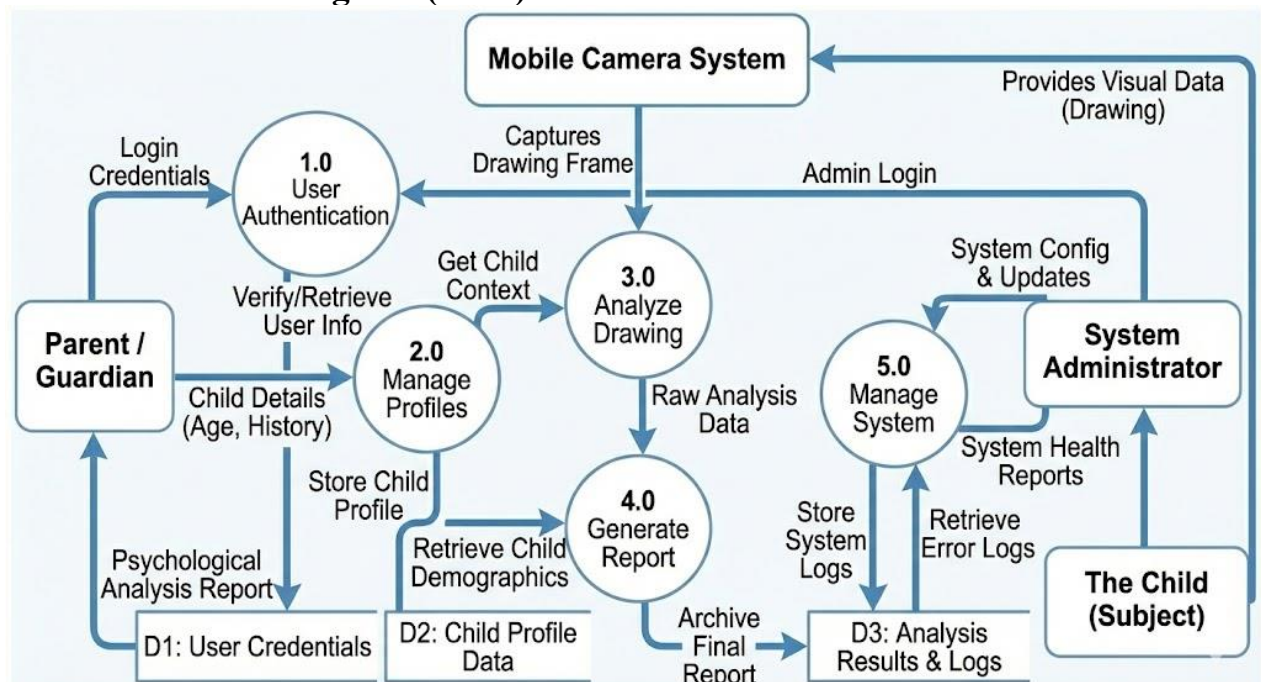


Figure 10: Data Flow Diagram

4.4 Entity Relationship Diagram (ERD):

The Entity Relationship Diagram defines the main entities of the Draw Mind AI system and their relationships, ensuring structured storage of user data, drawings, and emotional analysis results.

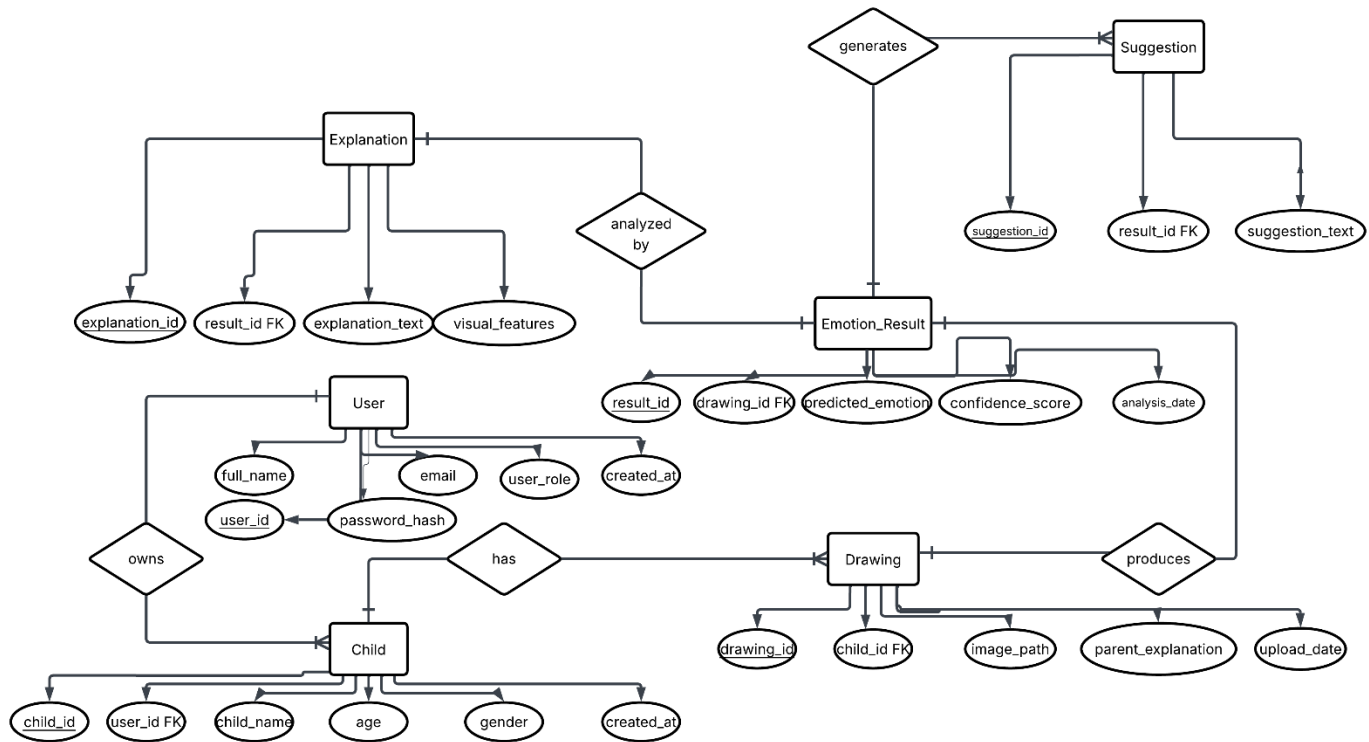


Figure 11: Entity Relationship Diagram

The database design of the Draw Mind AI system is derived directly from the Entity Relationship Diagram and defines the logical structure of data storage required to support system functionality. The database schema is implemented using PostgreSQL, where primary and foreign key constraints are enforced to maintain referential integrity.

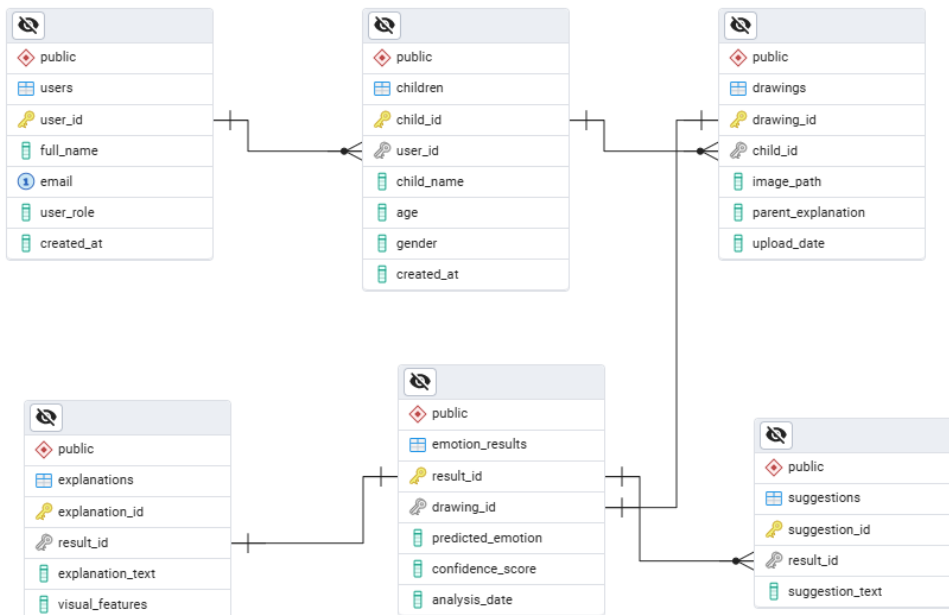


Figure 12: Database Scheme

1. **User Table:** Stores system users such as parents, caregivers, and teachers.
 - **user_id (Primary key):** Identifying each user
 - **full name:** Name of the user
 - **email:** User email address
 - **password_hash:** User's password encrypted
 - **user_role:** User role (parent, caregiver, teacher)

- **created_at**: Account creation date
- 2. **Child Table**: Stores information about children associated with users.
 - **child_id (Primary key)**: Identifying each child
 - **child_name**: Name of the child
 - **user_id (Foreign key)**: Referencing User
 - **age**: Child's age
 - **gender**: Child's gender
 - **created_at**: Record creation date
- 3. **Drawings Table**: Stores uploaded drawings for emotional analysis.
 - **drawing_id (Primary key)**: Identifying each drawing
 - **child_id (Foreign key)**: Referencing Child table
 - **image_path** : Path or reference to the stored image
 - **parent_explanation**: Additional information provided by the parent or caregiver about the drawing
 - **upload_date**: Date of upload
- 4. **Emotion Result Table**: Stores emotional analysis results for drawings.
 - **result_id (Primary key)**: Uniquely identifies each analysis result
 - **drawing_id (Foreign key)**: Referencing Drawing
 - **predicted_emotion**: Detected emotional category
 - **confidence_score**: Confidence level of the predicted emotion
 - **analysis_date**: Date of analysis
- 5. **Explanation Table**: Stores explainable AI outputs related to emotion results.
 - **explanation_id (Primary key)**: Identifying the explanation
 - **result_id (Foreign key)**: Referencing EmotionResult
 - **explanation_text**: Textual explanation of the emotional prediction
 - **visual_features**: Extracted visual features used in the analysis
- 6. **Suggestion Table**: Stores system-generated recommendations.
 - **suggestion_id (Primary key)**: Identifying the explanation
 - **result_id (Foreign key)**: Referencing EmotionResult
 - **suggestion_text**: Recommended guidance or actions

4.5 UML Use Case Diagram:

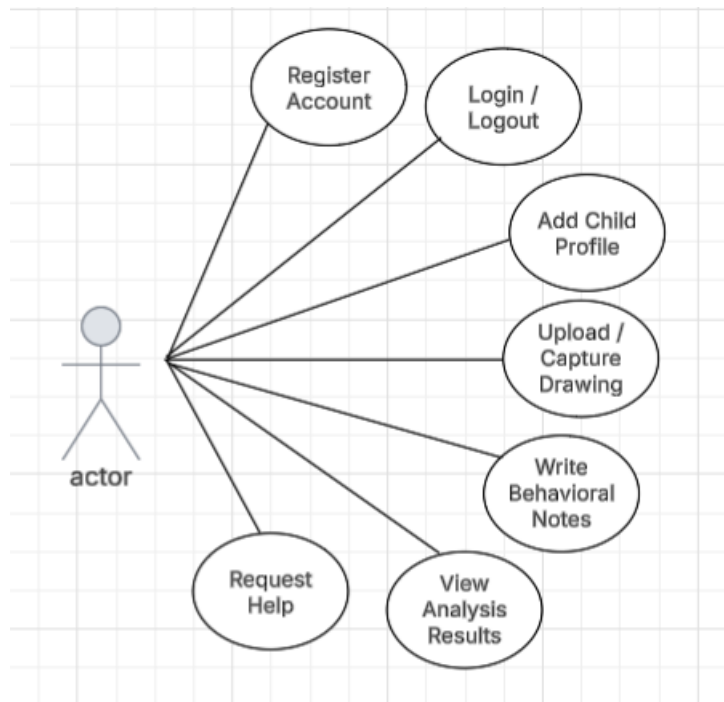


Figure 13: UML Use Case Diagram (Actor)

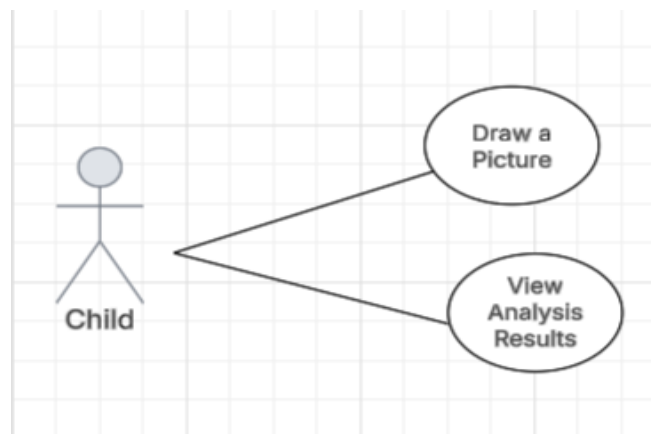


Figure 14: UML Use Case Diagram (Child)

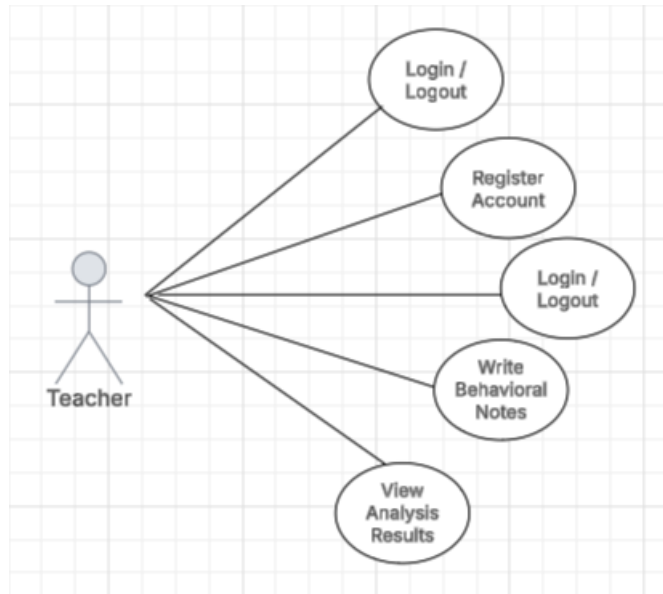


Figure 15: UML Use Case Diagram (Teacher)

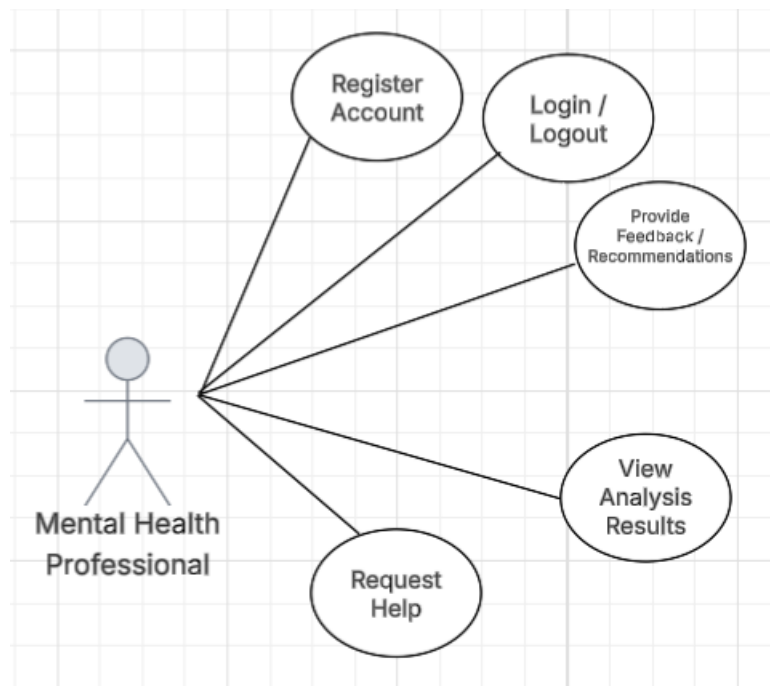


Figure 16: UML Use Case Diagram (Professionals)

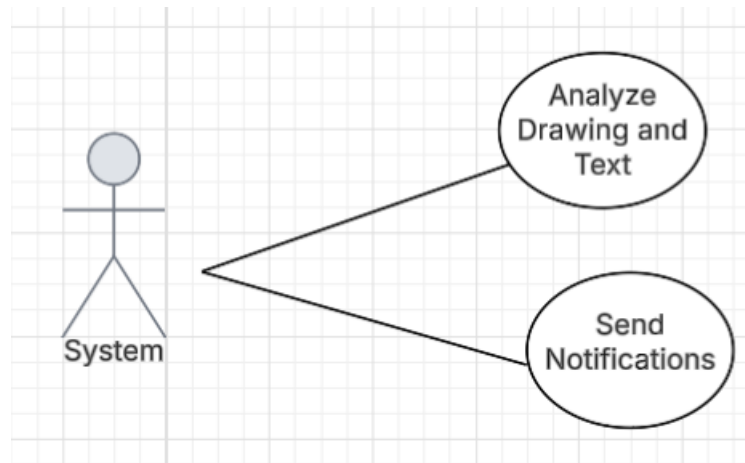


Figure 17: UML Use Case Diagram (System)

4.6 UML Sequence Diagram:

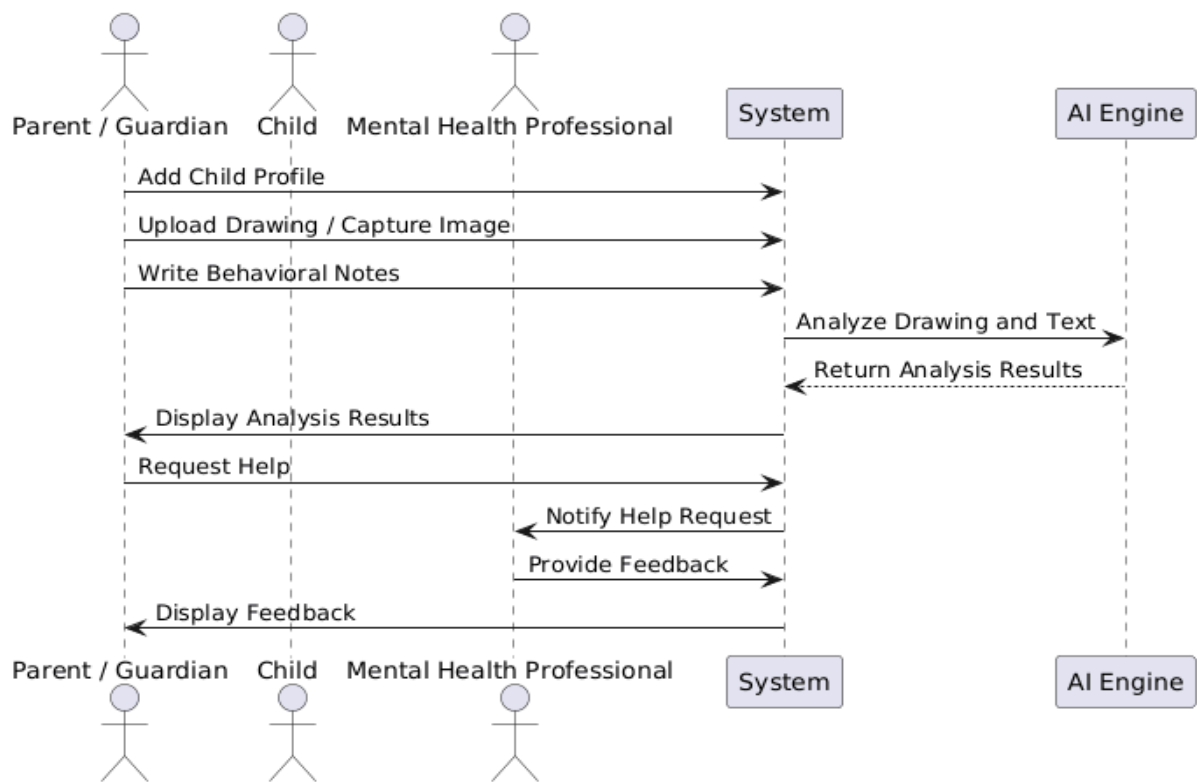


Figure 18: UML Sequence Diagram

4.7 UML Class Diagram:

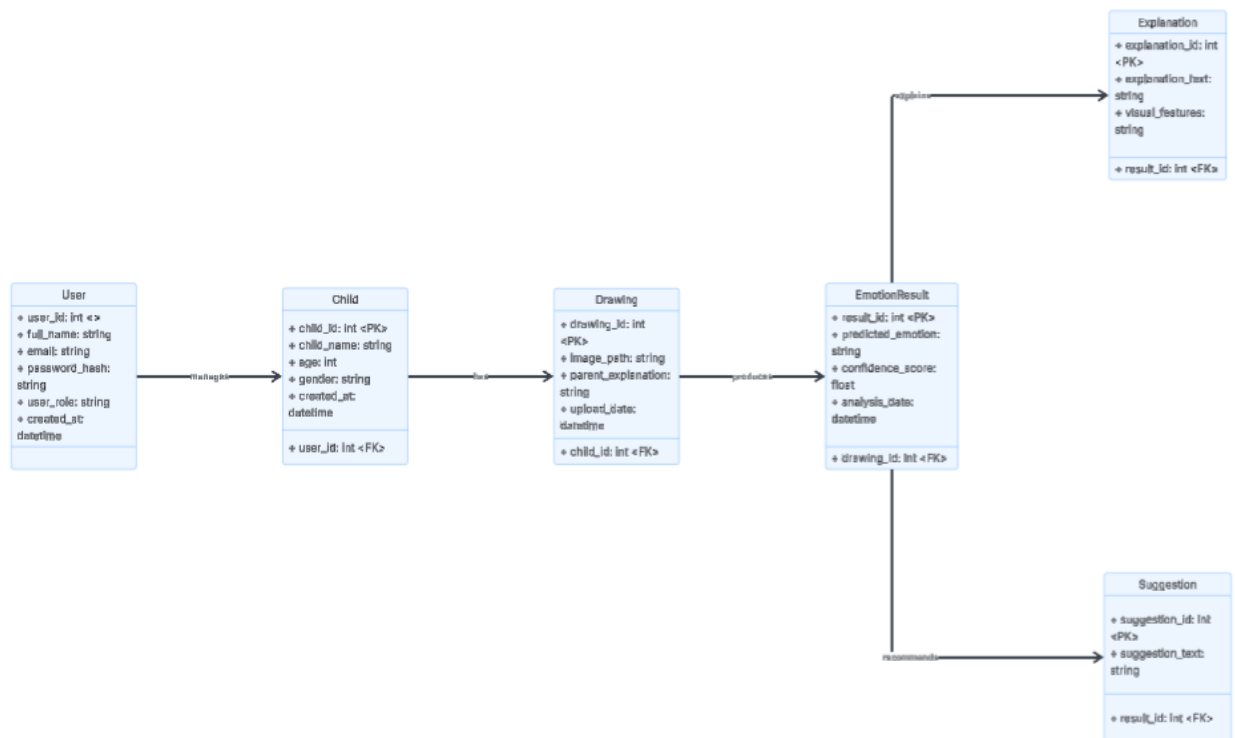


Figure 19: UML Class Diagram

4.8 Graphical User Interface:

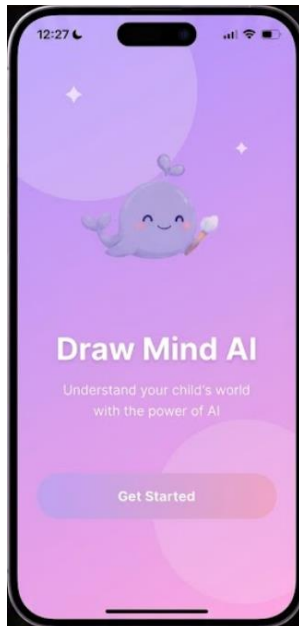
4.8.1 Overview

The graphical user interface is designed to be simple and user-friendly, with a focus on clear navigation and easy access to all system functions. It uses calm colors and clear visual elements, making it suitable for children and providing a smooth and comfortable user experience.

4.8.2 Main Screens

The application consists of several key screens, each designed to support a specific functionality:

-Welcome Screen:



- 1- Displays the application logo (Draw Mind AI)
- 2- Serves as the initial entry point to the app.

- Login Screen:



- 1- Allows existing users to log into their account securely.
- 2- Provides access to the application features and saved data.

Sign Up Screen:



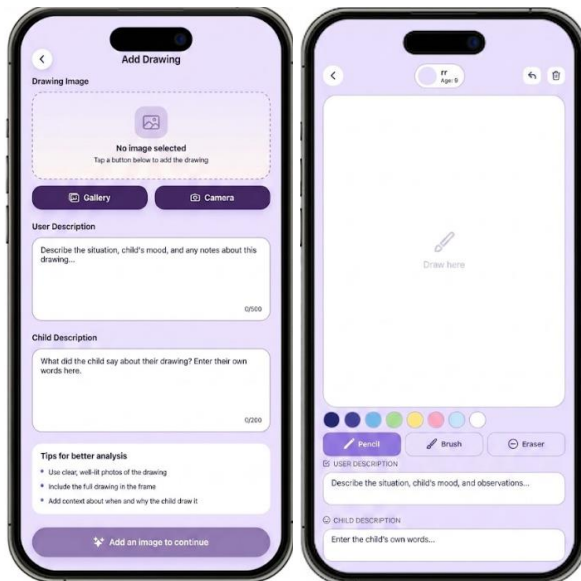
- 1-Allows new users to create an account in the application
- 2- Collects basic user information to set up a new profile including the user roles (parent, teacher, etc.)

-Home screen :



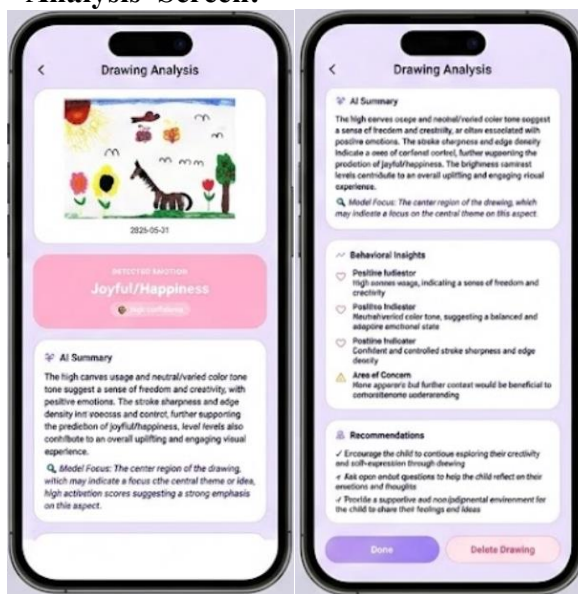
- 1- Provides an overview of each child's emotional state.
- 2- Supports adding as many children as the user wants.
- 3- Allows uploading or creating drawings within the application.

- Drawing Upload or Drawing Screen:



- 1- Allows users to upload drawings or enables the child to create new drawings easily, while also allowing parents to add their observations and the child to write their feelings or describe the drawing.
- 2- Initiates the AI analysis process.

- Analysis Screen:



- 1-Displays the detected emotional state of the child's drawing along with AI explanation.
- 2- Presents behavioral insights based on drawing features such as colors, strokes, and canvas usage.
- 3- Provides recommendations to support the child's emotional and creative development.
- 4- Allows users to save the results to the child's profile and delete analyzed drawings.

- Children Screen:



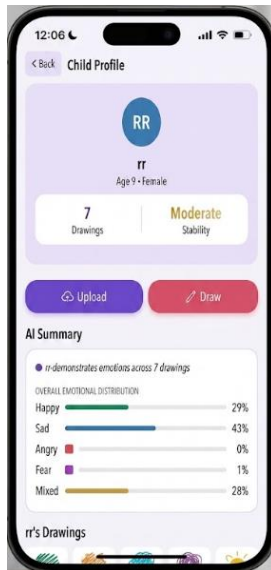
- 1- Displays a list of all children profiles added by the user
- 2- Allows the user to view each child's basic information and emotional overview.
- 3- Enables adding a new child profile to the system.

- Add Child Screen:

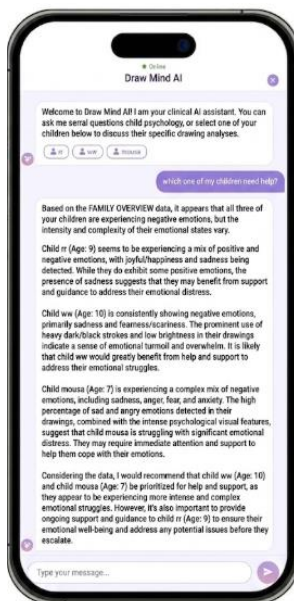


- 1- Allows users to add a new child profile.
- 2- Includes fields for child name, age, gender, and favorite activities.

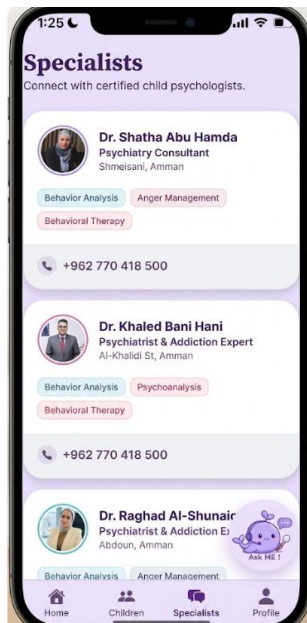
- Child Profile Screen:



- 1- Displays all previously analyzed drawings for a specific child.
 - 2- Shows detailed results and allows deletion of drawings and their analysis results.
 - 3- Includes a visual chart showing the distribution of emotional states over time (e.g., percentages of Happy, Sad, Angry, Fear).
- **Chatbot AI Assistant Screen:**

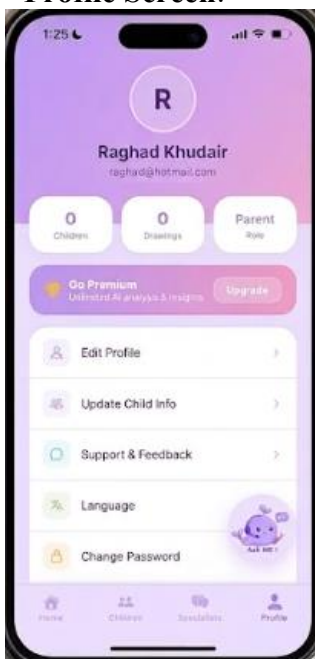


- 1- Enables users to ask questions related to their child's emotional state
 - 2- Provides explanations and guidance based on the AI analysis.
- **Specialists Screen:**



- 1- Provides contact information of qualified professionals.
- 2- Includes details about doctors' experience and specialization.
- 3- guides users to contact them externally.

- Profile Screen:



- 1- Allows users to manage their personal information .
- 2- Enables editing and updating child details
- 3- includes support, feedback, privacy policy, and terms of use.

4.8.3 User Flow

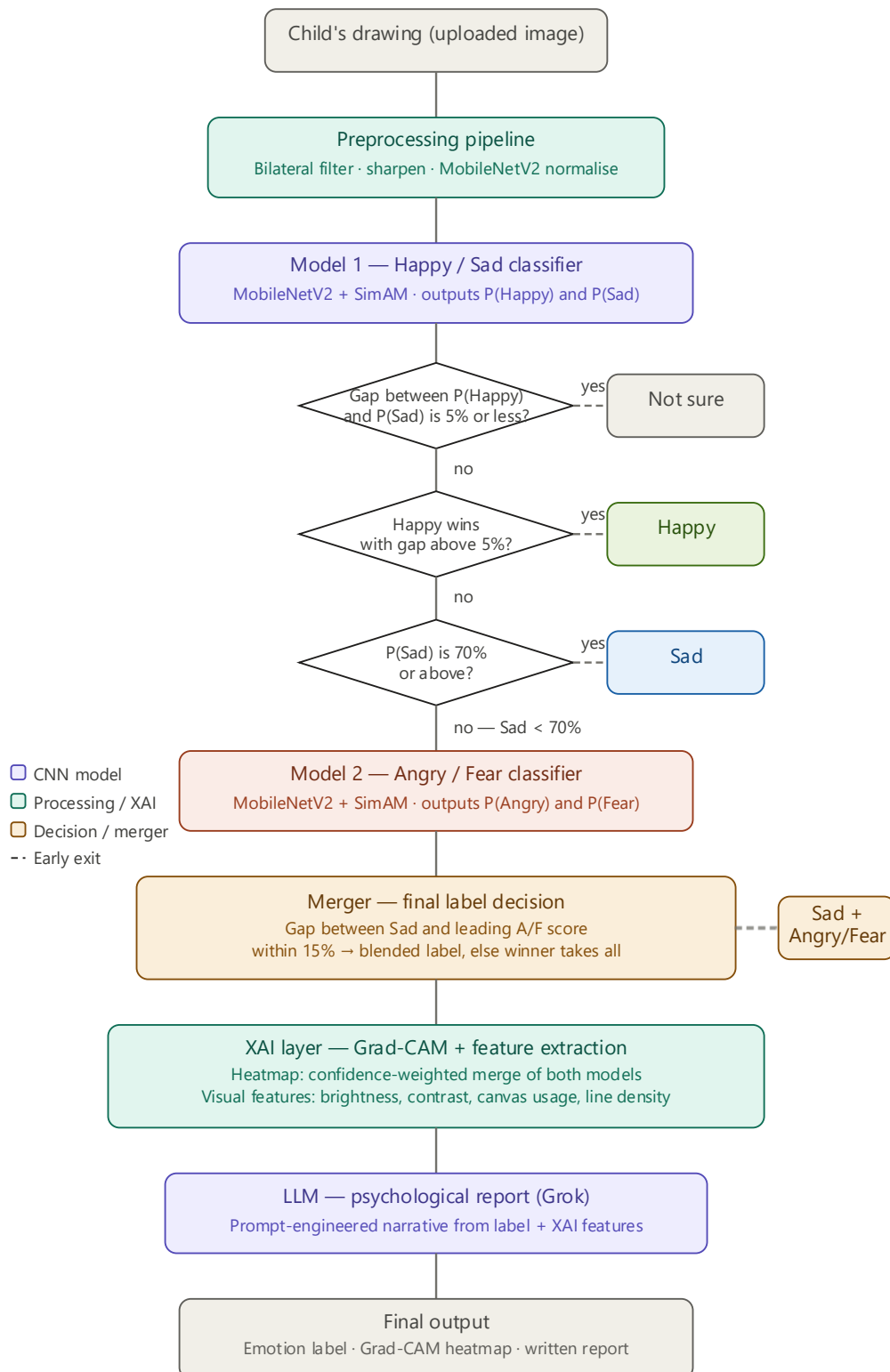
The system follows a clear and structured user flow.

- The user first sees the Welcome Screen, then proceeds to the Login Screen or Sign Up Screen to access or create an account.
- Then it opens the Home Screen, which provides an overview of all children and allows adding or selecting a child
- In Home Screen, the user has two options: Upload or Draw, after selecting one of these options, the user selects a child.

- The Upload or Draw screen will open, where the user can either upload a drawing or allow the child to create one using the drawing tool. Once the image is added, the user can enter the child's current emotional state in a dedicated box.
- The result is displayed in the Results Screen in a simple and clear format, and it is automatically saved and linked to the child's profile.
- The Children Screen displays all child profiles and allows the user to select an existing child profile or add a new one.
- When the user selects Add Child, the Add Child Screen opens, where they can enter the child's information
- When a specific child is selected, the Child Profile Screen opens to display previous drawings, analysis results, and includes a visual chart showing the distribution of emotional states over time (e.g., percentages of Happy, Sad, Angry, Fear), with the option to delete drawings and their results.
- The system also provides an AI Assistant Screen, where users can ask questions related to the child's emotional state and receive guidance.
- The Doctors Information Screen allows users to view contact details and information about specialists for further consultation.
- The Profile Screen enables users to manage their personal information, update child details, and access support and feedback options.

4.9 AI System Design :

Figure 20: AI System design



5. SYSTEM IMPLEMENTATION:

5.1 Introduction :

This chapter details the actual transformation of the DrawMind AI system design into a fully functional operational tool. System implementation marks the phase where logical specifications are converted into working code, encompassing the training of deep learning models, the establishment of a robust backend infrastructure, and the development of the client-side mobile interface.

Specifically, this section documents the specific configurations used to train the hierarchical vision models (Happy/Sad and Angry/Fear) using transfer learning with MobileNetV2 and the custom SimAM attention layer. It further outlines the backend development using FastAPI and PostgreSQL, detailing how deterministic computer vision outputs are integrated with the Llama-3.3 Large Language Model via advanced prompt engineering to produce explainable psychological interpretations. Finally, it presents the graphical user interface implementation, showcasing how complex analytical data is translated into a supportive, user-centric mobile experience.

5.2 Models Implementation:

5.2.1 Happy/Sad Model — Architecture and Training:

The Happy/Sad classifier was built using MobileNetV2 as a pre-trained backbone, leveraging weights from ImageNet to benefit from transfer learning on a relatively small dataset. The base model's convolutional layers were frozen during initial training to preserve the learned feature representations, while a custom classification head was added on top. This head consisted of a Global Average Pooling layer followed by a Dense layer of 128 units with ReLU activation and L2 regularization ($3e-4$), a Dropout layer with rate 0.3, a second Dense layer of 64 units with ReLU activation and L2 regularization, a second Dropout layer with rate 0.2, and a final sigmoid output neuron for binary classification.

Training was conducted in two distinct phases. In Phase 1, the backbone was fully frozen and the classification head was trained using the Adam optimizer with an initial learning rate of $1e-3$ and a cosine annealing schedule over 40 epochs, with plain Binary Cross-entropy loss and early stopping monitoring validation accuracy with a patience of 8 epochs. Online data augmentation was applied during training through a custom generator that randomly applied one of four transformations per image — original, horizontal flip, rotation $+15^\circ$, and rotation -15° — combined with random brightness and contrast jitter.

In Phase 2, the last 45 layers of the MobileNetV2 backbone were unfrozen for fine-tuning, while all Batch Normalization layers were kept frozen to preserve stable feature statistics during domain adaptation. The loss function was switched to Binary Focal Cross-entropy with $\gamma=2.0$, $\alpha=0.5$, and label smoothing of 0.05 to handle class imbalance more effectively. The learning rate was reduced to $1e-5$ with a cosine schedule ranging between $1e-7$ and $3e-6$ over 60 epochs, monitored by validation loss with patience of 8. The model was trained on the KIDO Children's Drawing Dataset with an 80/10/10 stratified split for train, validation, and test. With Test-Time Augmentation applied at inference using 5 image variants, the final model achieved a test accuracy of 73%, saved as `MobileNet_happy_sadfull_73acc.keras`.

Experiment	Configuration	Train	Val	Test	Notes
		Acc	Acc	Acc	
Baseline	MobileNetV2, no attention, BCE loss, LR= $1e-3$	~61%	~60%	~65%	Starting point, backbone frozen

+ Fine-tuning	Unfroze last 30	~75%	~67%	~70%	Fine-tuning with low LR=1e-5
Phase 2	backbone layers, BN				
	frozen + cosine learning				
	rate+ augmentation				
Without	min_lr=1e-7	~78%	~69%	~70%	
Cosine					
Learning					
Rate					
+ Random	Flip, rotation $\pm 15^\circ$,	~84%	~70%	~70%	
noise and	brightness/contrast				
zoom added	jitter+CosineLR				
Final Model	All above + Test-Time	~85%	~73%	73%	Saved as
+ SimAM	Augmentation (5 variants)				MobileNet_happy_sadfull_73acc.keras

Table 15: Happy/Sad Model experiments

5.2.2 Angry/Fear Model — Architecture and Training:

The Angry/Fear classifier shares the same MobileNetV2 backbone architecture, SimAM attention mechanism, and two-phase training strategy as the Happy/Sad model, but with several key differences that reflect the distinct characteristics of its training data and the stronger regularization requirements of a more visually complex emotional boundary.

The classification head uses the same structure — GlobalAveragePooling2D, SimAM attention, Dense(128, ReLU, L2=3e-4), Dense(64, ReLU, L2=3e-4), sigmoid output — but with significantly higher Dropout rates of 0.5 and 0.4 respectively, compared to 0.3 and 0.2 in the Happy/Sad model. This stronger regularization was necessary because the Angry/Fear dataset is smaller and the visual features separating anger from fear in children's drawings can be subtle, making the model more prone to overfitting without aggressive dropout.

The dataset was constructed by merging two sources: the Kaggle children's drawings dataset by Vishmi Perera and a custom Roboflow dataset collected specifically for this project, containing only the Angry and Fear classes. Exact duplicate images were identified and removed using MD5 hashing before splitting, resulting in a clean merged dataset split 80/10/10 stratified by class. Phase 1 used Adam with learning rate 5e-4, plain Binary Crossentropy, and a cosine schedule over 40 epochs. The augmentation generator applied 6 transformation types per image — original, horizontal flip, vertical flip, both flips combined, rotation +15°, and rotation -15° — plus brightness and contrast jitter, providing stronger augmentation coverage than the Happy/Sad model's 4-type strategy.

Phase 2 fine-tuned only the last 30 backbone layers compared to 45 in the Happy/Sad model, keeping Batch Normalization frozen throughout. The decision to fine-tune fewer layers reflects the smaller dataset size — unfreezing more layers on limited data risks catastrophic forgetting of the pretrained ImageNet features that the model relies on for generalization. The loss switched to Binary Focal Cross-entropy (gamma=2.0, alpha=0.5, label_smoothing=0.05), and the learning rate ranged between 1e-7 and 3e-6 via cosine schedule over 100 epochs. With Test-Time Augmentation using 8 variants at inference — including rotations at $\pm 10^\circ$ and $\pm 20^\circ$ and brightness scaling at 1.1x and 0.9x — the final model achieved a test accuracy of 93.41%, saved as 93_acc.keras.

Experiment	Configuration	Train Acc	Val Acc	Test Acc	Notes
Baseline	MobileNetV2, no attention, BCE loss, LR=5e-4	—	—	~80%	Single Kaggle dataset only
+ Dataset Merging	Combined Kaggle + Roboflow datasets, duplicate removal	—	—	~85%	Deduplication via MD5 hashing
+ SimAM Attention	MobileNetV2 + SimAM, higher dropout (0.5/0.4)	—	—	~89%	Stronger regularisation than HS model
+ Augmentation	6-way augmentation (flip H/V, rotation $\pm 15^\circ$, brightness)	—	—	~91%	More augmentation types than HS model
+ Fine-tuning Phase 2	Unfroze last 30 backbone layers, Focal loss, LR=3e-6	—	—	~92%	Tighter fine-tuning window
Final Model + TTA	All above + TTA (8 variants)	~96%	~93%	93%	Saved as 93_acc.keras

Table 16:: Angry/Fear Model experiments

5.2.3 SimAM Attention Mechanism Implementation:

SimAM (Simple, Parameter-free Attention Module) is a lightweight attention mechanism that computes a 3D attention weight for each neuron in the feature map without requiring any additional parameters. Unlike channel-based attention mechanisms such as SE-Net or spatial attention mechanisms like CBAM, SimAM operates directly on the feature tensor by computing an energy function based on the local spatial variance of activations.

The mechanism works by first computing the spatial mean across the height and width dimensions of the feature map. It then calculates the squared difference between each activation and the mean, which is normalized by the local variance plus a small regularization constant ($e_lambda = 1e - 4$). A sigmoid function is then applied to produce the final attention weights, which are multiplied elementwise with the original feature map. This allows the model to suppress irrelevant background regions and amplify the discriminative stroke patterns that carry emotional information.

The SimAM layer was implemented as a custom Keras layer decorated with `tf.keras.utils.register_keras_serializable()` to ensure it could be saved and reloaded correctly during deployment. The implementation required careful handling during model serialization, as custom layers must be explicitly registered before any `load_model` call.

5.2.4 Model Architecture Comparison — Why MobileNetV2 + SimAM Was Chosen:

Several architecture choices were considered for this project, including ResNet50, EfficientNetB0, ConvNet and DenseNet. MobileNetV2 was selected as the primary backbone for the following reasons. First, its depth wise separable convolutions make it significantly more lightweight than alternatives such as ResNet50 or DenseNet, which is essential given the limited size of the training datasets and the need for deployment in a resource-constrained local server environment. Second, MobileNetV2 includes inverted residual blocks with linear bottlenecks that preserve gradient flow during training, which proved beneficial given the relatively small number of training samples available.

SimAM was chosen over other attention mechanisms specifically because it introduces zero additional trainable parameters. Given that the datasets used in this project contain a few thousand samples at most, adding parameterized attention modules such as CBAM would risk overfitting. SimAM achieves the same spatial attention effect through a purely mathematical energy function, making it well-suited to this data-scarce setting.

1.2.5 Hierarchical Architecture:

Rather than training a single four-class classifier, the system implements a hierarchical decision architecture that chains the two binary classifiers in a gated sequence. This design choice was motivated by the observation that the emotional spectrum can be naturally divided into two groups — positive emotions (Happy) and negative emotions (Sad, Angry, Fear) — and that a single four-class model would need to learn all four decision boundaries simultaneously, which proved difficult given the dataset sizes and the visual similarity between certain emotion pairs. The hierarchical pipeline operates through a strict sequence of gated decisions. Every input image is first passed through the Happy/Sad model. If the absolute gap between the Happy and Sad probabilities is 5% or less, the system immediately returns "Not Sure" without invoking the second model, acknowledging that the first model has insufficient confidence to make a reliable prediction. If Happy wins with a clear gap above 5%, the system returns Happy immediately and the second model is never called. If Sad dominates with a probability of 70% or above, the system also returns Sad directly without consulting the Angry/Fear model, as the confidence level is considered high enough to skip the second stage. Only when Sad wins with a probability below 70% and a gap above 5% does the system invoke the Angry/Fear model.

Once the Angry/Fear model runs, the final decision compares the Sad probability against the maximum of the Angry and Fear probabilities. If the gap between Sad and the leading AF emotion is within 15%, the system returns a blended label — "Sad + Angry" or "Sad + Fear" — to reflect genuine emotional ambiguity in the drawing. If the Sad probability leads by more than

15%, Sad wins alone. If the Angry/Fear model is clearly more confident than Sad by more than 15%, that emotion is returned as the final output.

The complete set of decision thresholds used in the deployed system is summarized in the table above. This architecture provides two key advantages over a flat four-class approach. First, it reduces unnecessary computation by skipping the second model in the majority of cases where the first model is already confident. Second, it preserves the specialization of each binary model — each classifier was trained to distinguish only two emotions, which produces sharper and more reliable decision boundaries than a single model forced to separate all four simultaneously.

1.3 Explainable AI (XAI) Implementation:

This section describes how the DrawMind AI system integrates Explainable Artificial Intelligence (XAI) techniques to improve the transparency, interpretability, and reliability of the prediction process. Since emotion analysis based on children's drawings is a sensitive field related to psychological interpretation, the system was designed not only to provide prediction results but also to explain how these results are generated. Explainability plays an important role in increasing user trust, reducing the “black-box” nature associated with deep learning models, and enabling researchers, psychologists, and parents to better understand the mechanism the system uses to make its decisions. The system combines Grad-CAM heatmap visualization with handcrafted visual feature extraction to explain how children's drawings are analysed and how prediction decisions are made within the intelligent model.

5.3.1 Grad-CAM Heatmap Generation:

Grad-CAM is used to generate visual heatmaps that highlight the most important regions influencing the model's prediction. The system applies TensorFlow Gradient Tape on the MobileNetV2 backbone to compute activation maps from the final convolutional layer.

The generated heatmap shows where the model focused during prediction and is overlaid on the original drawing using OpenCV color mapping. Red regions indicate high attention, while blue regions represent lower importance. This visualization helps users understand which areas of the child's drawing contributed most to the emotional classification result.

Since Draw Mind AI uses two connected models, the system can merge heatmaps from both models using confidence-weighted averaging to produce a unified visual explanation. This combined heatmap provides a more comprehensive representation of the decision-making process across the cascaded architecture.

In addition, the system automatically generates a short textual interpretation describing the primary focus region and activation intensity. These explanations provide additional insight into the behavior of the AI model and improve the interpretability of the prediction process.

1. Original Drawing



2. AI Focus Area (Grad-CAM)



5.3.2 Visual Features of the Image:

In addition to Grad-CAM visualization, DrawMind AI extracts a set of handcrafted visual features directly from the drawing using classical image processing techniques. These features provide an

additional interpretable layer that supports the model's predictions and improves the transparency of the analysis process.

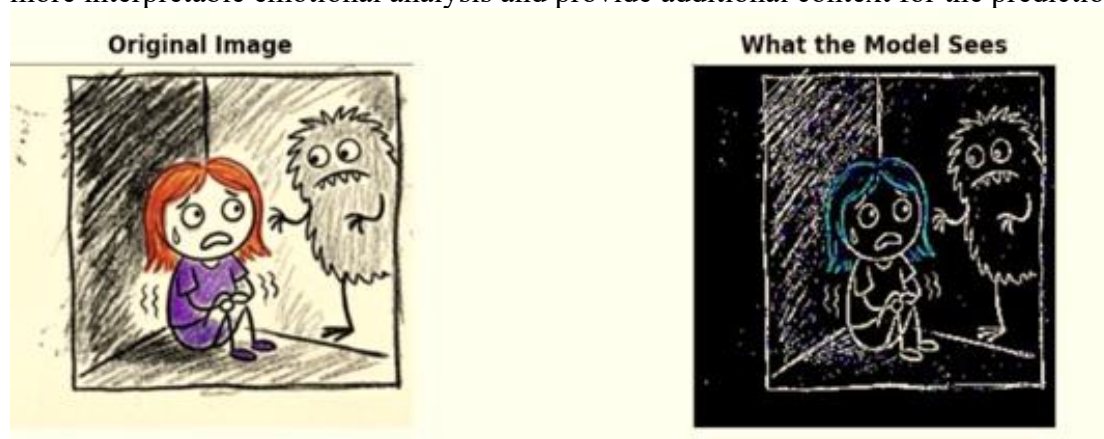
The `extract_features()` function computes several visual properties, including:

- Brightness
- Contrast
- Edge Density
- Canvas Usage
- Stroke Sharpness
- Dominant Color Tone

Each feature provides psychological indicators related to the drawing style and emotional expression. For example, darker drawings may reflect emotional intensity, while high edge density may indicate complex or anxious drawing behavior. Similarly, sharp angular strokes may be associated with tension or aggression, whereas smoother curves may suggest calmness or positive emotional states.

The system also classifies the dominant color tone into categories such as High Red Tones, Heavy Dark/Black Strokes, Very Light/Faint Strokes, or Neutral/Varied, based on predefined image analysis rules.

All extracted visual features are integrated into the final AI-generated report to support clearer, more interpretable emotional analysis and provide additional context for the prediction results



5.5 Backend Implementation:

The backend of Draw Mind AI is built using FastAPI, which was chosen because it handles asynchronous operations well-meaning the server can manage file uploads, database queries, and AI inference calls concurrently without one blocking another. The architecture is organized so that each of these responsibilities is handled separately: incoming requests are routed cleanly, uploaded drawing files are cached efficiently, and the AI pipeline is triggered in a controlled, sequential way once the necessary inputs are ready.

5.5.1 Database Implementation :

The database runs on a serverless PostgreSQL instance hosted through Neon's cloud platform. All the connection management, session handling, and schema mapping is done through SQLAlchemy, which keeps the database logic clean and consistent across the codebase.

The screenshot shows the Neon console interface. On the left, there's a sidebar with navigation options like Dashboard, Branches, Integrations, Settings, Overview, Monitoring, SQL Editor, Tables, Backup & Restore, Data Masking, Data API, Auth, and Collapse menu. The main area displays the 'production' table with the following data:

suggestion_id	result_id	suggestion_text	emotion_result
21	21	Encourage the child to...	emotion_result
22	21	Offer support and vali...	emotion_result
23	21	Consider exploring oth...	emotion_result
24	22	Encourage the child to...	emotion_result
25	22	Support the child in d...	emotion_result
26	22	Provide a safe and sur...	emotion_result
27	23	Encourage the child to...	emotion_result
28	23	Provide a supportive a...	emotion_result
29	23	Consider engaging the ...	emotion_result
39	27	Encourage the child to...	emotion_result
40	27	Explore the child's da...	emotion_result
41	27	Foster an open and sup...	emotion_result
42	28	Encourage the child to...	emotion_result
43	28	Engage the child in op...	emotion_result
44	28	Consider seeking the s...	emotion_result
45	29	Encourage the child to...	emotion_result
46	29	Explore the child's re...	emotion_result
47	29	Support the child in d...	emotion_result

Looking at `database.py`, the connection is set up with a couple of important safety measures. The engine uses `pool_pre_ping=True`, which means it checks that the database is actually reachable before attempting to run a query — this prevents the server from sending requests into a dead connection and crashing silently. On top of that, `sslmode="require"` is enforced through the `connect_args`, so all traffic between the FastAPI server and the Neon database is encrypted in transit. Sessions themselves are handled through a generator function called `get_db()`, which yields an active database session and then closes it inside a `finally` block regardless of what happens during the request. This makes sure connections are always cleaned up properly and don't pile up into deadlocks or memory leaks over time.

The database schema is split across six tables, each handling a distinct part of the system's data:

users: Stores parent account records, registration details, and hashed passwords.

children: Holds child profile data — age, demographic attributes, profile icons — each linked to a parent account through a foreign key.

drawings: Stores the file paths pointing to uploaded images in the `static/drawings` directory, along with any text descriptions and the current processing status of each drawing.

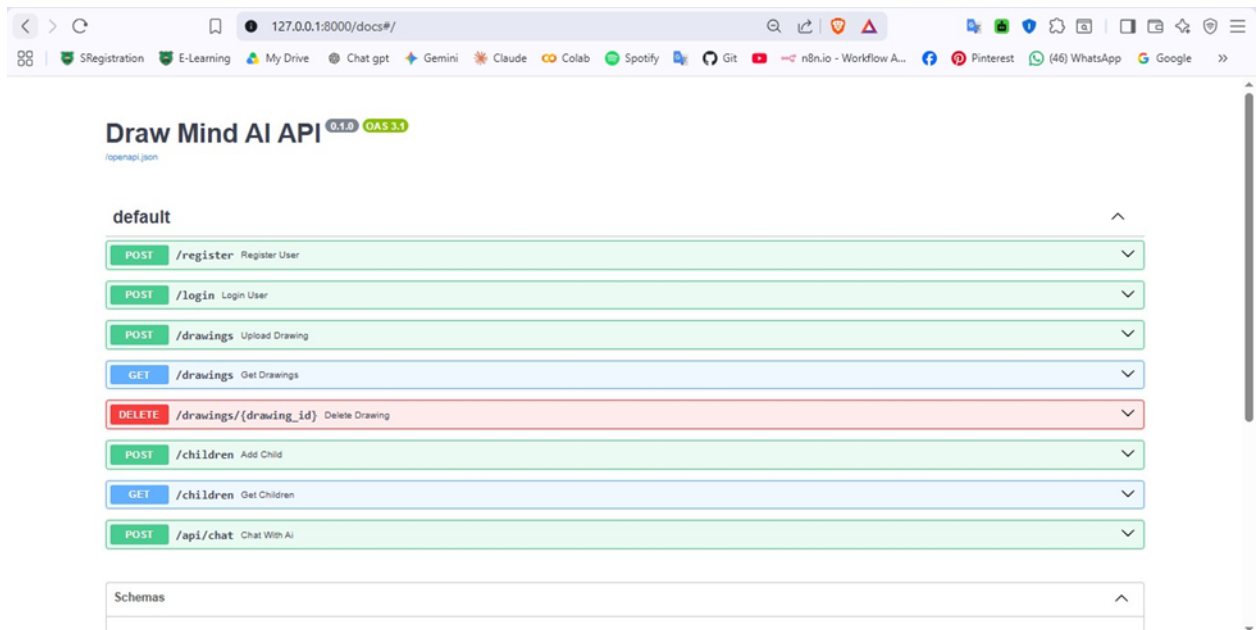
emotion_results: The core analytical output table, storing the detected emotion label, the model version that produced it, and the confidence score.

explanations: Holds the full psychological interpretation for each drawing — the visual feature breakdown, contextual analysis notes, and the narrative summary generated by the language model.

suggestions: Stores the list of parenting guidance recommendations, each linked back to a specific result row via a `result_id` foreign key. Cascade-delete is configured here so that if a result gets removed, its suggestions are automatically cleaned up too.

5.5.2 API and Server Integration:

All communication between the frontend and the database goes through a RESTful API defined in `main.py`. CORS is configured using `CORSMiddleware` with open settings, which allows the mobile client to make requests to the server regardless of origin — something that's necessary during development and for supporting multiple device types in production.



The API is organized into seven endpoints, each handling a specific part of the application's functionality:

- **POST /register & POST /login:** Handle user account creation and authentication. Passwords are hashed and verified using `CryptContext` with `bcrypt`, and login returns the user's profile data for the frontend to work with.
- **POST /children & GET /children:** Manage child profiles. The POST endpoint validates the incoming data and creates a new child record linked to the parent's account, while the GET endpoint retrieves all child profiles associated with a given parent email.
- **POST /drawings:** The most complex endpoint in the system. It receives an uploaded drawing file as a binary `UploadFile`, saves it to the `static/drawings` directory with an epoch-stamped filename to avoid collisions, then runs the full computer vision and deep learning pipeline synchronously — extracting the 14 visual features, running the emotion classifiers, calling the language model — and finally commits all the results across the relevant database tables in a single transaction.
- **GET /drawings:** Fetches a child's full drawing history for the dashboard. It uses SQLAlchemy's `joinedload` strategy to pull data from all the related tables — drawings, emotion results, explanations, and suggestions — in a single database round-trip rather than making separate queries for each, which keeps response times fast.
- **DELETE /drawings/{drawing_id}:** Removes a drawing record from the database. Because cascade-delete is configured on the related tables, this single operation automatically cleans up all associated emotion results, explanations, and suggestions without needing separate delete calls.
- **POST /api/chat:** Handles the conversational AI feature, taking the user's message along with the stored session history and generating a contextual response based on the child's profile and past drawing results.

5.6 Psychological Feature Extraction and LLM Integration:

The analytical core of Draw Mind AI does more than just classify an image into an emotion category. Instead of stopping at a label, the system combines the numerical outputs from the computer vision and deep learning pipeline with a large language model — specifically Llama-3.3-70b-versatile, accessed through the Groq API — to produce a full psychological interpretation. The idea is that the deterministic, offline calculations handle the objective measurement side of things, while the language model handles turning those measurements into something meaningful and readable for a parent.

5.6.1 Extracted Visual Features:

Before anything gets sent to the language model, the system first runs a local preprocessing pipeline inside `ai_helpers.py` to pull a set of numerical features directly from the drawing image. These features are specifically chosen to reflect the kinds of visual cues a psychologist would look at during a projective drawing assessment — things like how much of the page the child used, how heavy or light their strokes are, and what colors dominate the drawing.

The extraction is done using OpenCV, with a different algorithm handling each feature:

- **Canvas Spatial Usage (`canvas_usage`):** An inverse binary threshold (`cv2.threshold`) is applied to separate the drawn areas from the blank canvas, then `cv2.findContours` is used to find the outer boundaries of the drawing. The bounding rectangle around those contours is measured against the total canvas size to get a percentage of how much space the child filled.
- **Stroke Continuity and Sharpness (`edge_density` & `sharpness_ratio`):** Canny edge detection (`cv2.Canny`) picks up high-frequency pixel changes across the image — essentially finding where lines and edges are — while `cv2.goodFeaturesToTrack` identifies localized corners and points of geometric change. Together these give a sense of how sharp, continuous, or erratic the child's strokes are.
- **Color Tone and Dominance (`dominant_tone`):** The image is split into its R, G, and B channels using `cv2.split`, and the distribution of pixel values across each channel is analyzed. The system specifically looks for heavy concentrations of dark pixels (grayscale below 50) and high-saturation reds, both of which are associated in child psychology literature with emotional intensity and behavioral arousal.
- **Brightness and Contrast (`brightness` & `contrast`):** The image is converted to grayscale using `cv2.cvtColor`, and then the mean and standard deviation of the pixel values are computed across the whole canvas, giving a general measure of how light or dark and how varied the drawing is overall.

Once all the visual features are extracted, the preprocessed image is passed through the deep learning pipeline inside `AI_Service` in `ai_service.py`. The first model (`happy_sad_compat.h5`) — which includes SimAM self-attention layers to help it focus on the most emotionally relevant parts of the image — outputs probability scores for happy and sad. If the gap between those two scores is small enough to fall below the confidence threshold (`NOT_SURE_THRESHOLD = 5.0`), the system doesn't just pick the higher one and move on. Instead, it automatically routes the image to a second specialized model (`angry_fear_compat.h5`) to check whether the ambiguity is actually pointing to something more specific, like anger, fear, or anxiety.

5.6. AI Prompt Engineering:

Once the visual features and classifier probabilities are ready, everything gets packaged together inside the `call_groq` function and sent to the language model as a structured system prompt. The prompt is carefully engineered to do a few specific things: keep the output grounded in the actual visual data, prevent the model from generating confident-sounding clinical claims it has no basis for, strip out any technical machine learning terminology that would confuse a parent, and ensure the response always comes back as a valid, parseable JSON object.

The exact prompt template used to govern the language model is as follows:

SYSTEM PROMPT SPECIFICATION CONTAINER (Ingested within call_groq Environment)

ROLE DESIGNATION:

- Act exclusively as an expert clinical child psychologist and pedagogical counselor.

EMPIRICAL INPUT CONTEXT BLOCK:

- Final Detected Structural Emotion Matrix: {friendly_emotion}
- Raw Neural Classifier Probabilities:
Happy: {Happy}% | Sad: {Sad}% | Angry: {Angry}% | Fear: {Fear}%
- Automated Feature Ingestion Matrix:
 - * Spatial Canvas Density Profile: {canvas_usage}%
 - * Chromatic Stroke Dominance Profile : {dominant_tone}
 - * Edge Geometric Continuity Score: {sharpness_ratio} | Density: {edge_density}%
 - * Grayscale Mean Brightness/Contrast: {brightness} | {contrast}

CAREGIVER BEHAVIORAL META-DATA LOGS:

- Subject Demographic Profile: Age: {age}
- Primary Caregiver Context Notes: "{parent_text}"
- Child Declarative Explanation: "{child_text}"

MANDATORY EXECUTION CONSTRAINTS:

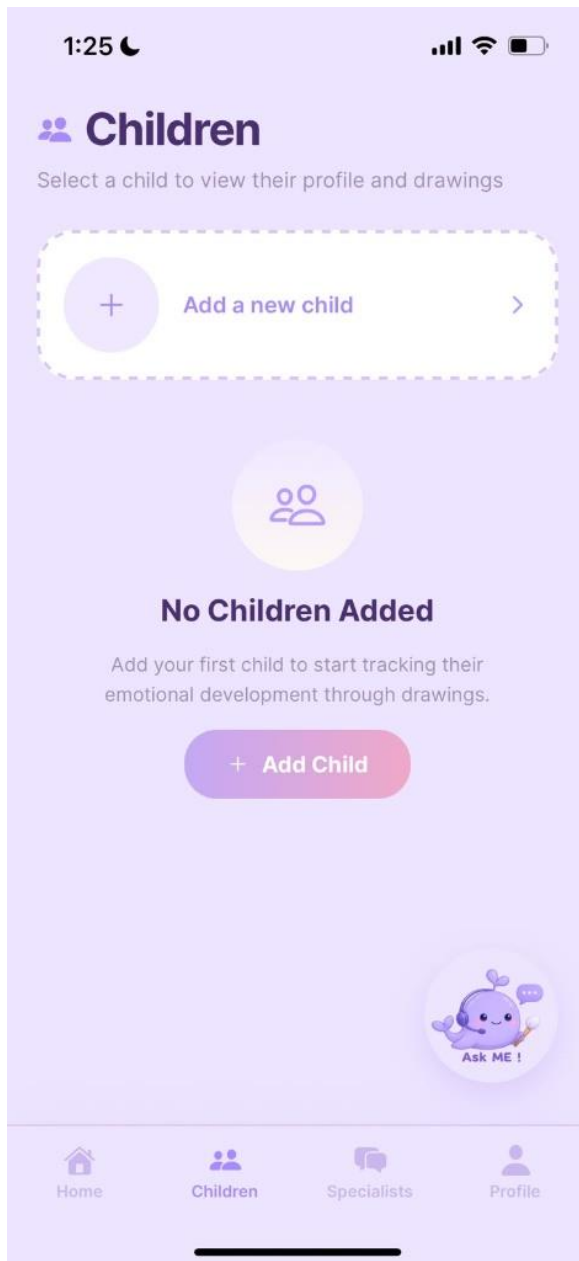
1. Write a dedicated segment titled "Contextual Analysis" that explicitly links the caregiver context notes and child descriptions to the 14 extracted visual variables.
2. Rely strictly on the psychology of spatial usage, color distribution, and stroke textures.
3. Suppress all technical deep learning model jargon and machine learning baseline metrics.
4. Deliver supportive, actionable, non-clinical pedagogical guidance tailored for parents.
5. Format the output strictly as a parseable JSON object.

5.7 Graphical User Interface Implementation (High-Fidelity Prototype):

The interface is basically the part of the Draw Mind AI system that takes all the complicated data and psychological analysis happening in the background and turns it into something a regular person can actually use. Since this is a mobile app, the design was built around three main things: managing client settings, tracking a child's development, and giving counselors a way to communicate through the app.

5.7.1 Interface Layout and Demographic Monitoring Screens

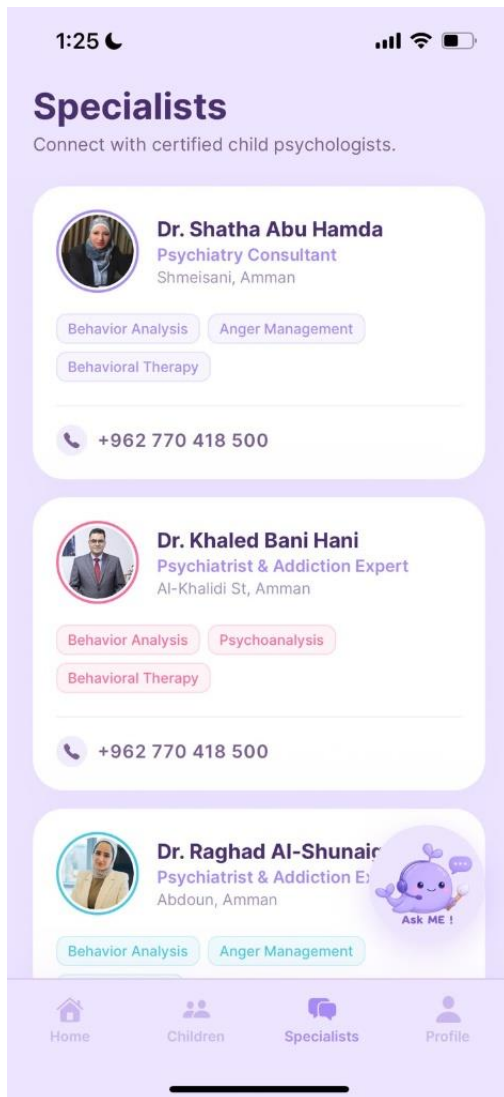
The main goal here was to keep things simple, especially since the people using this app aren't necessarily tech-savvy. To make navigation easier, there's a menu bar fixed at the bottom of the screen so users can always find their way around.



As shown in the image, if no children have been added yet, the app doesn't just show a blank screen it displays a helpful message and a clear "+ Add Child" button to get the user started. Once tapped, a form pops up where parents can enter their child's details, which then gets saved to the database after going through some basic validation.

5.7.2 Mental Health Resources and Specialist Directory

Beyond the analytics side of things, the app also has a directory screen where users can find real mental health professionals, which is a nice touch because it connects the automated insights to actual human support.



As shown in the image, specialists are listed as profile cards, each showing the professional's name, their areas of focus (like Behavior Analysis, Anger Management, or Behavioral Therapy), and a direct phone number. The example in the screenshot shows professionals based in Amman, Jordan. There's also a floating button with the app's mascot and an "Ask ME!" label that stays visible no matter what screen the user is on, so parents can always get back to the AI assistant whenever they need it.

6.SYSTEM TESTING AND EVALUATION:

6.1 Introduction:

This chapter presents the system testing and evaluation of the DrawMind AI framework. The main objective of this phase is to assess the correctness, robustness, and overall performance of the developed system, and to verify that it satisfies both functional and non-functional requirements defined in the previous chapters.

The evaluation process is structured into multiple levels of testing, including unit testing, integration testing, and usability testing, to ensure that each system component functions correctly both independently and as part of the complete pipeline. In addition, a comparative analysis of the experimental results is provided to evaluate the performance of the proposed AI models, including the Happy/Sad and Angry/Fear classifiers, as well as the overall hierarchical architecture.

Furthermore, this chapter discusses the effectiveness of the explainable AI (XAI) components, the reliability of the prediction outputs, and the system's ability to handle real-world drawing inputs. The results obtained from testing demonstrate that DrawMind AI provides a stable, interpretable, and accurate framework for emotional analysis of children's drawings, with improved transparency through Grad-CAM visualization and handcrafted feature-based explanations.

6.2 Requirements Validation and Completeness:

This section validates whether the non-functional requirements specified during the system analysis phase were successfully achieved in the final implementation of DrawMind AI. The evaluation process focused on testing the system's overall performance, usability, reliability, security, explainability, and AI prediction quality to ensure that the developed application satisfies the intended project objectives.

The system demonstrated good performance by processing uploaded drawings and generating emotional analysis results within a reasonable response time while maintaining efficient resource usage. In terms of scalability, modular system architecture supports future expansion, including additional users, child profiles, and larger datasets without major architectural modifications.

To ensure security and privacy, user accounts, uploaded drawings, and emotional analysis results are securely managed within the system database, while authentication mechanisms help prevent unauthorized access. Reliability and availability were also considered during testing, where the application maintained stable operation with minimal failures and provided appropriate error handling during invalid uploads or network interruptions.

The maintainability requirement was addressed through the use of a modular software structure that separates AI models, image processing components, explainability modules, and frontend functionalities. This structure simplifies future updates, debugging, and system enhancements. The AI models were evaluated to verify prediction accuracy and consistency using trained datasets of children's drawings. In addition, usability testing confirmed that the interface is simple and user-friendly for non-technical users such as parents and psychologists, allowing easy interaction with the analysis dashboard and prediction reports.

Furthermore, ethical and transparency requirements were integrated into the system by clearly informing users that the results generated are AI-based emotional indicators rather than medical diagnoses. The explainable AI (XAI) module also improves transparency by providing Grad-CAM visual explanations and interpretable visual feature analysis to support user trust and understanding of the prediction process.

Finally, the personalization requirement was achieved through support for multiple child profiles, where each child has separate drawing history, prediction records, and emotional analysis reports. Overall, the testing and evaluation process confirmed that DrawMind AI satisfies the majority of the specified non-functional requirements and provides a reliable, interpretable, and user-friendly intelligent analysis system.

6.3 System Testing:

To actually test whether the Draw Mind AI system works the way it's supposed to, a full round of testing was carried out. The testing was broken down into three main areas: unit testing to check individual parts of the code, integration testing to make sure everything works together, and usability testing to see how real caregivers interact with the app.

6.3.1 Unit Testing:

Unit testing is basically about checking that each individual piece of the code works correctly on its own, without worrying about how it connects to everything else. This includes things like helper functions, logic routines, and database connection handlers — each one gets tested with specific inputs to make sure it gives back the right output.

Table below summarizes the unit tests that were run and what the results looked like:

Unit ID	Module / Function Tested	Input Used	Expected Output	Actual Output	Result
UT-01	database.py / get_db()	Opening a database session	Active connection is returned and properly closed after use	Session opened and closed correctly	✓ Pass
UT-02	ai_helpers.py / canvas_usage	A drawing covering 35% of the canvas	Returns 35.0 as a float value	Correct value returned	✓ Pass
UT-03	ai_helpers.py / dominant_tone	An image filled mostly with red	Identifies and returns the dominant warm color	Red tone correctly detected	✓ Pass
UT-04	ai_service.py / Happy/Sad CNN	A drawing classified as clearly happy	High confidence score toward the "happy" category	Probability scores correctly calculated and passed to the rest of the system	✓ Pass

Table 17: Unit Testing

```

Mindful-AI-Companion > artifacts > mobile > app > login.tsx > LoginScreen
161 export
304 > 8000 Aa ab .* No results
PROBLEMS 3 TERMINAL OUTPUT ... uvicorn - fastapi-server + v
(venv) PS C:\Users\user\OneDrive\Attachments\Desktop\DrawMindAI\Mindful-AI-Comp
panion\Mindful-AI-Companion\artifacts\fastapi-server> uvicorn main:app --reload
INFO: 127.0.0.1:60967 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60984 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60891 - "GET /drawings?child_id=11 HTTP/1.1" 200 OK
File saved to static\drawings\child_14_1780218690.png. Starting AI analysis.
..
Starting Analysis for: static\drawings\child_14_1780218690.png
Emotion: Sad (82.6%)
DEBUG: Data sent to Groq -> Parent: '..', Child: '..'
[✓] Analysis Complete!
[✓] AI analysis finished. Saving to database...
INFO: 127.0.0.1:61020 - "POST /drawings HTTP/1.1" 200 OK
INFO: 127.0.0.1:61033 - "GET /static/drawings/child_14_1780218690.png HTTP/
1.1" 200 OK
INFO: 127.0.0.1:61035 - "GET /drawings?child_id=14 HTTP/1.1" 200 OK

```

when a drawing is uploaded, the system runs through each step one by one and logs the result. It confirms that the file was saved successfully, shows the emotion detected by the AI (for example, "Sad" with 82.6% confidence), logs that the data was sent off for further analysis, and completes the whole process without any crashes or errors.

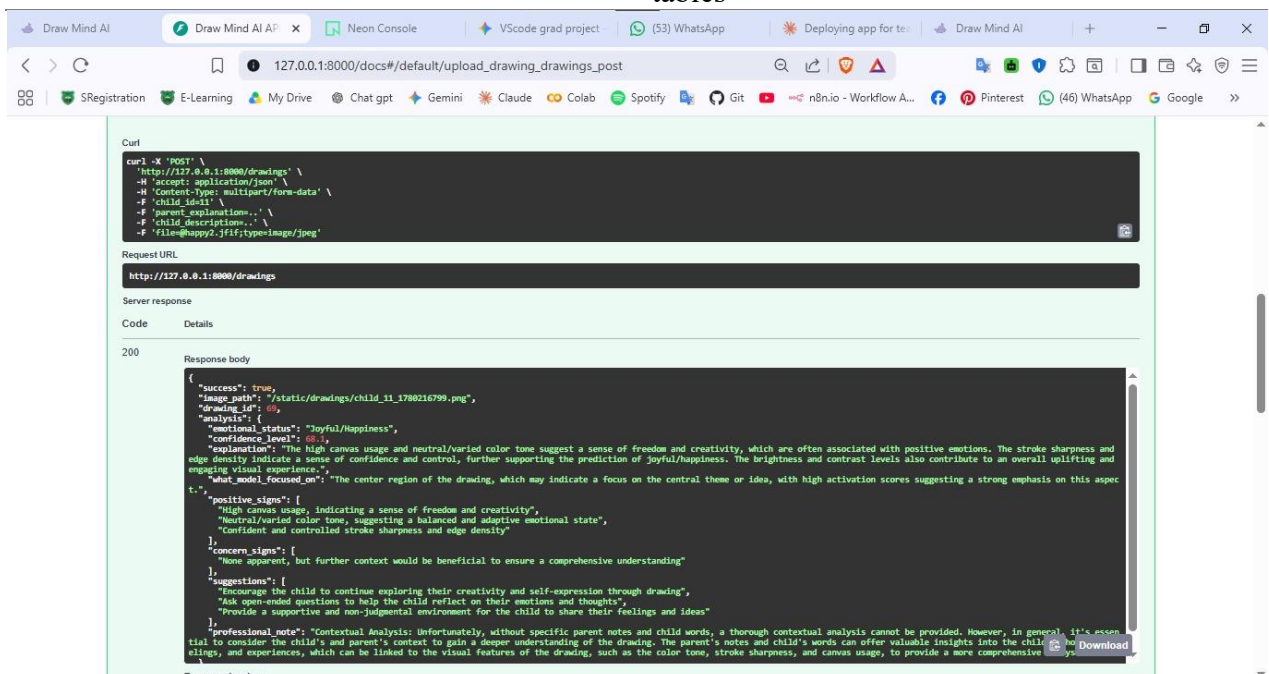
6.3.2 Integration Testing:

Integration testing is about making sure all the different parts of the system actually work together properly, not just on their own. This includes checking that the API endpoints save files correctly, pass data through to the AI models, communicate with the external language model, and store everything in the database without anything breaking along the way.

Table below covers the main integration tests that were run:

Integration Testing Results					
integration ID	Components Tested	Scenario	Expected Result	Actual Result	Result
IT-01	FastAPI Server ↔ passlib	Logging in with the	Request is rejected with a 400 error	Access correctly denied	Pass

IT-02	File Upload ↔ OpenCV	wrong password Uploading an image with parent details	File is saved and 14 image features are extracted	Image saved and features passed to the next step	Pass
IT-03	CNN Models ↔ Groq LLM	Sending emotion scores to the AI for analysis	A proper response is returned from the LLM	Groq returned a valid response	Pass
IT-04	LLM Output ↔ Neon Database	Saving the AI's response to the database.	Data is validated and inserted into the right tables	All data committed to the database successfully	Pass



As shown in the image, sending a drawing to the `/drawings` endpoint came back with a 200 success code. The response included the drawing's database ID (`drawing_id: 69`), the detected emotion, a short summary, and a list of suggestions for parents — all returned cleanly with no errors or delays.

6.3.3 Usability Testing:

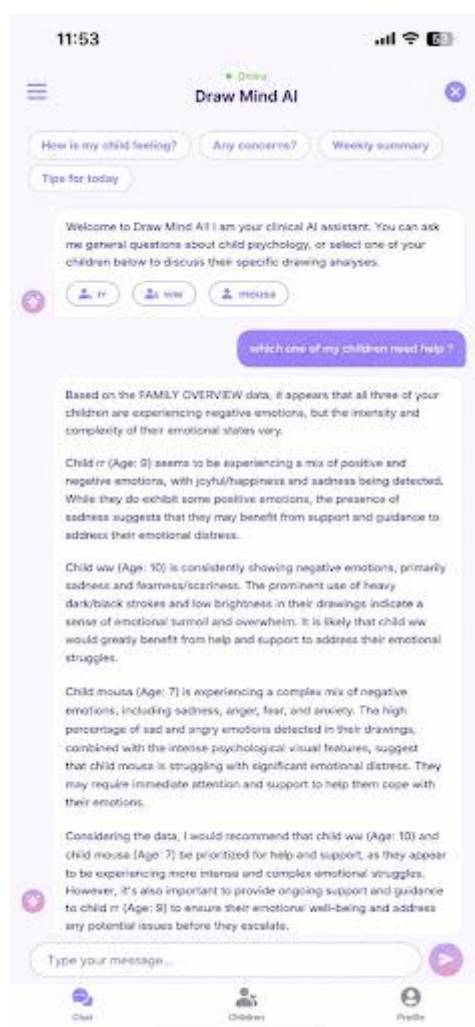
Usability testing was about putting the app in front of the people who will use it mainly parents and educators and seeing how they get on with it. The goal was to check that someone with no technical background can still figure out how to register a child, read the results, and understand the suggestions without feeling lost or overwhelmed.

Table summarizes the usability tests and what was observed:

Usability ID	Screen Tested	What the User Was Asked to Do	What Was Being Evaluated	What Happened	Benchmark
UT-01	Child Profiles Dashboard	Add a new child through the	How easy the form is to understand and	Parents filled in the form quickly and got	Met

UT-02	Specialist Directory	empty-state prompt Browse local therapist profiles and try calling one	fill out within 15 seconds How readable the profiles are and whether the call button works	instant feedback on any mistakes Profiles were easy to read and the phone links opened the dialer correctly	Met
UT-03	Chat Assistant	Ask a general question about how their children are doing	Whether the advice is easy to understand and free of technical language	The assistant grouped the children's data and gave clear, practical parenting tips	Met

Table 18: Usability testing results



As shown in the image, when a parent typed something like "which one of my children needs help?", the app responded with a simple breakdown for each child — in this case, rr, ww, and Mousa. It pulled out things like the child's age and drawing patterns (such as heavy dark strokes or low brightness) and turned them into straightforward, caring advice. Nothing felt too clinical or hard to follow, and the layout held up fine without anything breaking or becoming hard to read.

6.4 Comparative Summary of ALL Experiments:

Model Name	Test Accuracy	Precision (Macro Avg)	Recall (Macro Avg)	F1-Score
EfficientNetV2-S	70.53%	0.71	0.71	0.7156
ConvNextTiny	70.73% (<i>Val</i>)	0.71	0.71	0.7030
EfficientNetB1	70.16%	0.70	0.70	0.7000
DenseNet	69.73%	0.70	0.70	0.6954
EfficientNetB0	69.24%	0.70	0.68	0.7268*
<u>MobileNetV2</u>	<u>~0.73%</u>	<u>~0.73</u>	<u>~0.73</u>	<u>~0.73</u>

Table 19: Comparative summary of all models

Phase 1: Happy vs. Sad Model Evaluation and Selection During the experimental phase, we rigorously evaluated several lightweight convolutional neural network architectures—including variants of Efficient Net, DenseNet, and ConvNext—to establish a baseline for our primary emotion classification task (Happiness vs. Sadness). Initial experiments demonstrated a highly consistent performance ceiling across the board, with most architectures converging near a 70% accuracy rate. EfficientNetV2-S provided a highly stable baseline, achieving a test accuracy of 70.53% alongside symmetric precision and recall scores (0.71).

However, the breakthrough in our experiments was achieved using MobileNetV2. By integrating custom architectural enhancements—specifically the SimAM (Similarity-based Attention Module)—and optimizing the feature extraction pipeline, MobileNetV2 successfully broke through the previous performance bottleneck. It achieved the highest overall performance with a peak accuracy of 73.00%. This superior accuracy, combined with MobileNetV2's exceptionally lightweight footprint and rapid inference speed, made it the definitive choice for the Tier-1 "Gatekeeper" model in our final cascading backend architecture.

Phase 2: Angry and Fear Unlike the Happy/Sad model, for which a large number of architectures, regularization strategies, and training configurations were explored and compared — as reflected in the detailed experimental table in the previous section — the Angry/Fear model was subjected to only two configurations before a final architecture was selected. This was not an oversight but a deliberate response to the fundamental constraints of the underlying dataset. The combined Angry/Fear dataset, constructed by merging two external sources and removing 123 duplicate images through MD5 cryptographic hashing, yielded only 910 clean images in total — roughly one-twelfth the size of the KIDO dataset used to train the Happy/Sad model. Even after applying a stronger six-way augmentation strategy covering flipping, rotation, zoom, shift, shearing, and brightness and contrast jitter, the effective training pool remained too limited to support the kind of broad architectural experimentation carried out for the Happy/Sad model. Running multiple training cycles over such a small dataset would have produced unreliable comparative conclusions, since minor random variation in weight initialization or augmentation sampling could shift accuracy figures by several percentage points independently of any architectural change.

Beyond raw size, the quality and nature of the data itself posed additional barriers. The Angry/Fear dataset was assembled from two sources that were not originally designed for emotional classification of children's drawings: one required manual labelling of images that had no prior emotion annotation, and the other provided drawings from a different domain whose labels carry inherent subjectivity — a single drawing depicting a dark, imposing figure could legitimately represent either anger or fear depending on the child's intent. This label noise, combined with the well-documented psychological overlap between the two high-arousal emotions, means the dataset contains a level of ambiguity that additional training experiments alone cannot resolve. Furthermore, as the threshold analysis in section 6.4.3 confirms, the Angry/Fear model was deliberately positioned as a secondary classifier in the hierarchical pipeline — one that is only invoked when the Happy/Sad model returns a low-confidence Sad prediction. Given this supporting role, the project prioritized achieving a reliable and well-regularized solution within the available data rather than exhaustive exploration of alternatives that the dataset size could not meaningfully distinguish between.

6.4 Base Model Experimental Results:

6.4.1 Happy/Sad Model — Detailed Results:

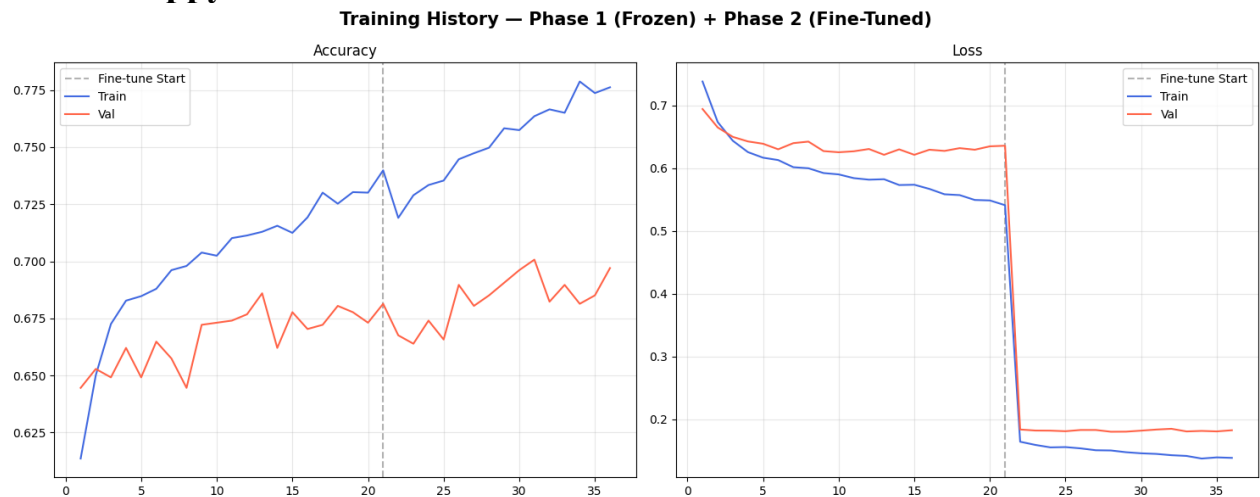


Figure 21: Happy/Sad acc and loss curves

The training curves for the Happy/Sad model reveal a notably more challenging learning pattern compared to the Angry/Fear model, which directly reflects the greater difficulty of distinguishing happiness from sadness in children's drawings. In Phase 1 with the frozen backbone, the training accuracy climbed gradually from approximately 62% to around 73% by epoch 21, while the validation accuracy fluctuated considerably between 64% and 73% throughout, never stabilizing into a smooth curve. This high variance in validation accuracy across Phase 1 indicates that the Happy/Sad boundary is genuinely ambiguous and that the model required more training signal to find reliable discriminative features. In Phase 2, fine-tuning triggered an immediate and sharp drop in both train and validation loss — from approximately 0.63 to below 0.2 — while the training accuracy continued rising toward 78%. However, the validation accuracy remained lower than the training accuracy and showed continued fluctuation, peaking around 70% with occasional drops. This persistent train-val gap suggests that despite regularisation, the model found it harder to generalise the Happy/Sad boundary to unseen drawings than the Angry/Fear model, which is consistent with the final test accuracy of 73%.

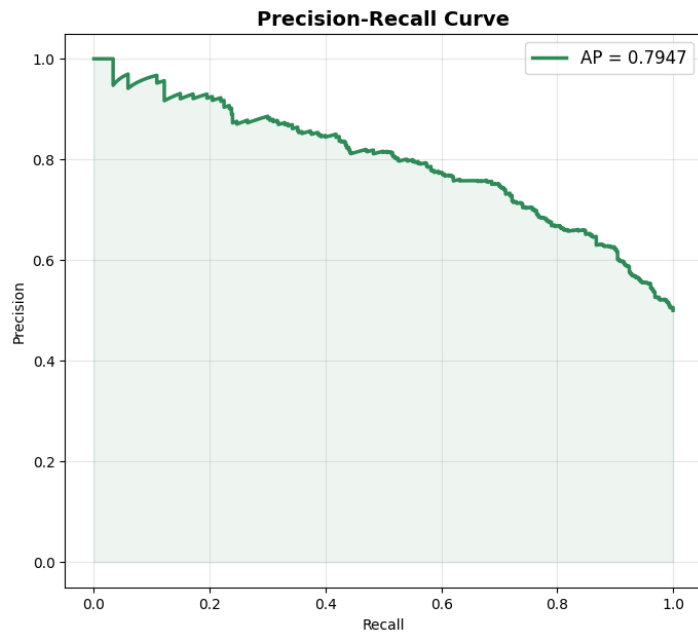


Figure 22: Angry/Fear Precision-Recall curves

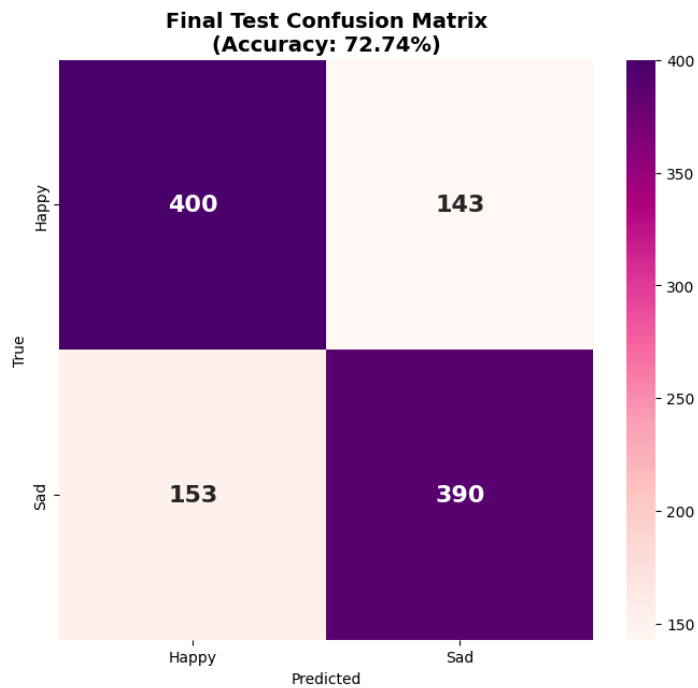


Figure 23: Happy/Sad Confusion Matrix

Class	precision	recall	f1-score	support
Happy	0.72	0.74	0.73	543
Sad	0.73	0.72	0.72	543

accuracy			0.73	1086
macro avg	0.73	0.73	0.73	1086
weighted avg	0.73	0.73	0.73	1086

Table 20: Happy/Sad Classification report

The confusion matrix reveals a balanced performance across both classes. The model correctly identified 74% of Happy drawings and 72% of Sad drawings, with misclassifications distributed roughly equally in both directions. This symmetry suggests the model did not develop a systematic bias toward either class, which is supported by the balanced class weights applied during training. The similar error rates for both classes reflect the genuine difficulty of the task — Happy and Sad children's drawings share many visual features such as similar stroke densities and figure proportions, making the boundary between the two classes inherently ambiguous.

6.4.2 Angry/Fear Model — Detailed Results:

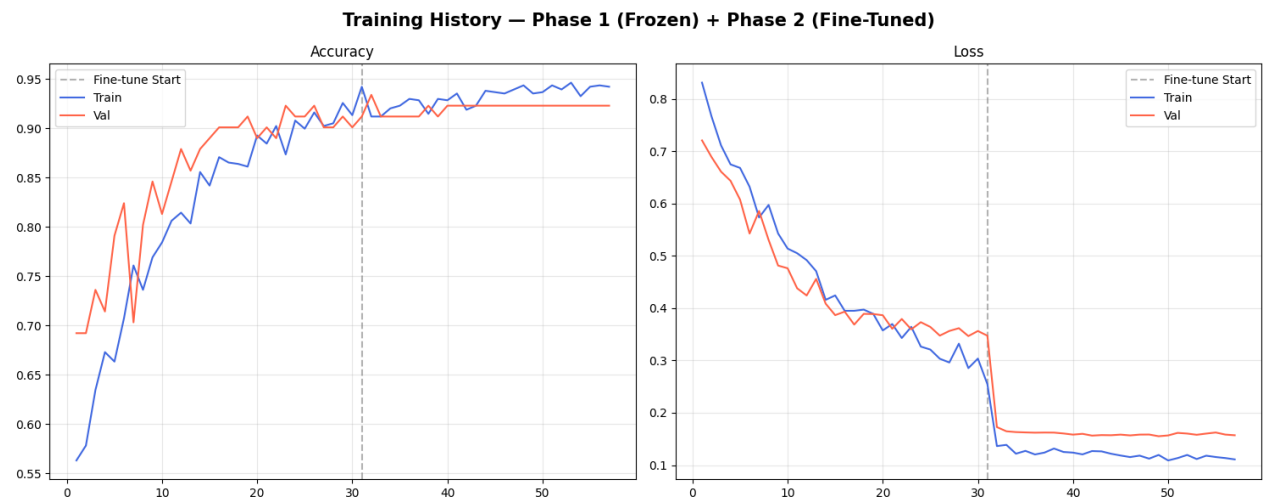


Figure 24: Angry/Fear acc and loss curves

The training curves for the Angry/Fear model demonstrate strong and stable learning across both phases. In Phase 1, with the backbone frozen, the validation accuracy rose sharply from approximately 69% in the first epoch to around 90% by epoch 10, notably leading the training accuracy in the early epochs — a sign that the pretrained ImageNet features transferred effectively to the drawing domain even before any fine-tuning. Both curves converged and stabilized around 90-92% by the end of Phase 1 at epoch 31. In Phase 2, fine-tuning of the last 30 backbone layers produced an immediate and significant drop in loss from approximately 0.35 to below 0.15, while validation accuracy stabilized at around 92-93%. The training accuracy continued to climb slightly above the validation accuracy in Phase 2, indicating a small degree of overfitting, but the gap remained narrow and the validation curve remained stable, confirming that the Dropout rates of 0.5 and 0.4 along with L2 regularization were effective in controlling it.

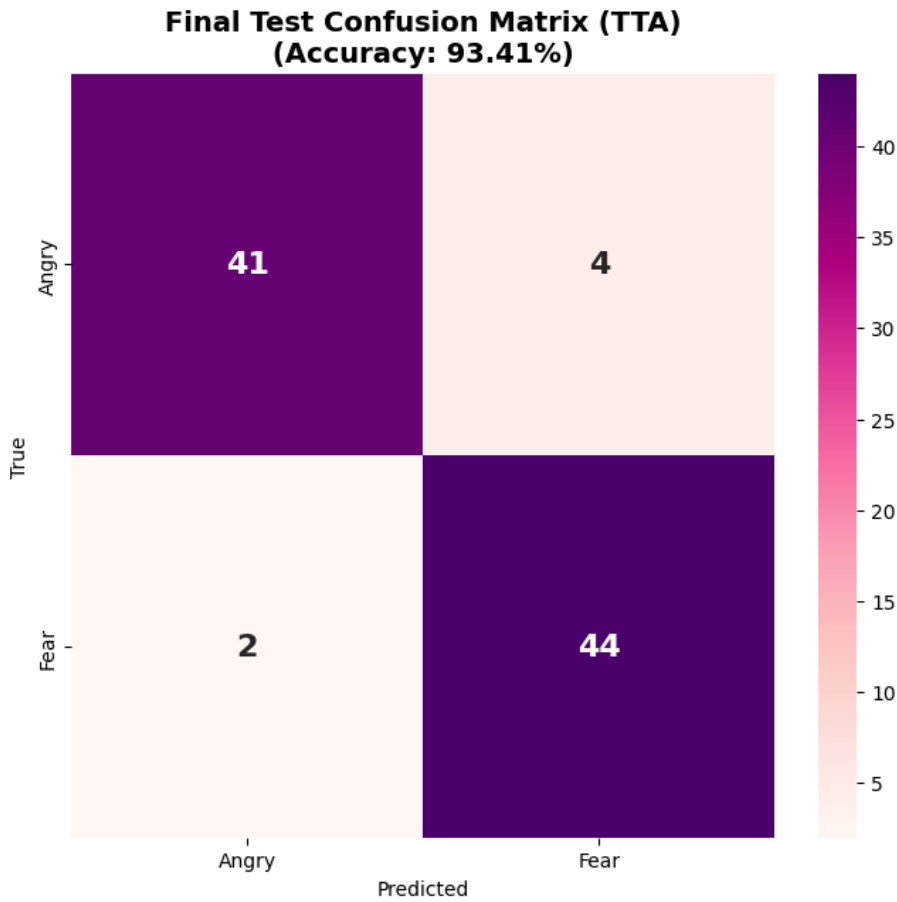


Figure 25: Angry/Fear Confusion Matrix

Class	Precision	Recall	F1-Score	Support
Angry	0.93	0.93	0.93	45
Fear	0.93	0.93	0.93	46
Accuracy			0.93	91
Macro Avg	0.93	0.93	0.93	91
Weighted Avg	0.93	0.93	0.93	91

Table 21:Angry/Fear Classification Report

The confusion matrix shows near-perfect classification performance for both classes. The model achieved 93% precision and 93% recall for both Angry and Fear, with misclassifications being rare and distributed equally between both directions. This strong performance is attributed to the combination of SimAM attention directing the model toward the most emotionally relevant regions of the drawing, the larger effective dataset from merging two sources, and the stronger augmentation strategy using 6 transformation types.

6.4.3 Hierarchical Architecture — Detailed Results:

The hierarchical decision system was evaluated on a set of children's drawings spanning all four emotion categories as well as ambiguous cases. Since the intermediate decision paths of the system were not individually logged during testing, the evaluation focuses on the overall system behaviour and the qualitative outcomes observed during testing.

The system demonstrated efficient gating behavior across test images. The majority of Happy drawings triggered the early exit path immediately after the first model, meaning the Angry/Fear model was never invoked for those cases, reducing inference time significantly. Drawings with clearly dominant Sad predictions above 70% also existed early without calling the second model. Only drawings where Sad won with moderate confidence required the full two-model pipeline, which represented a minority of test cases.

The blended emotion outputs — "Sad + Angry" and "Sad + Fear" — appeared most frequently in drawings that contained mixed visual signals, such as figures depicted with both drooping postures and aggressive stroke patterns, or scenes containing threatening elements alongside sad facial expressions. Rather than forcing a single incorrect label onto these ambiguous cases, the system's ability to output a blended label more accurately reflects the psychological reality that children's emotional states are not always singular or clearly defined.

The "Not Sure" output was triggered when the Happy/Sad gap fell within 5%, and in those ambiguous cases the Groq LLM was given the additional context of the GradCAM heatmap description and visual features to produce a more nuanced psychological interpretation, compensating for the model's uncertainty with qualitative reasoning.

Threshold Parameter	Value	Purpose
NOT_SURE_THRESHOLD	5%	Gap between Happy/Sad below which result is "Not Sure"
ABSOLUTE_SAD_THRESHOLD	70%	Sad confidence above which Model 2 is skipped
DIFFERENCE_THRESHOLD	15%	Sad vs AF gap above which Sad wins alone
SAD_AF_BLEND_THRESHOLD	15%	Gap below which a blended emotion is returned

Table 22: Thresholds of the system

The threshold values presented in the table above were not selected arbitrarily but were determined empirically through repeated testing on a diverse set of children's drawings across all four emotion categories. During the testing phase, it was observed that the Happy/Sad model consistently produced more confident and stable probability outputs compared to the Angry/Fear model. This behavior is directly explained by the difference in dataset size — the Happy/Sad model was trained on the KIDO dataset which is significantly larger and more diverse than the combined Angry/Fear dataset, giving it a stronger and more generalized feature representation. As a result, a lower Not Sure threshold of 5% was sufficient for the Happy/Sad model, since its predictions were rarely ambiguous. In contrast, the Angry/Fear model, being trained on a smaller and noisier dataset, required a more generous blending threshold of 15% to avoid overriding the Happy/Sad model's confident Sad predictions with weak Angry/Fear signals. The absolute Sad threshold of 70% was set to protect cases where the Happy/Sad model was highly confident in a Sad prediction, ensuring that the Angry/Fear model could only override Sad predictions when they were genuinely uncertain. Through this empirical calibration process, the threshold values were tuned to reflect the relative reliability of each model rather than treating both models as equally trustworthy.

6.4.3 Overview Summary:

Component	Task	Accuracy	Dataset Size	Notes
Happy/Sad Model	Binary: Happy vs Sad	73%	~10.000 images	Larger dataset, balanced classes
Angry/Fear Model	Binary: Angry vs Fear	93.41%	~730 images	Merged datasets, stronger augmentation
Hierarchical System	4-class + blended + Not Sure	—	-----	Gated cascade, no retraining needed

Table 23: Overview summary of the models

The overall system achieves strong performance by exploiting the specializations of each binary model rather than attempting to train a single model on all four classes simultaneously. The 93% accuracy of the Angry/Fear model and the stable 73% accuracy of the Happy/Sad model combine to produce a system that handles the full emotional spectrum reliably, with graceful degradation to blended or Not Sure outputs in genuinely ambiguous cases rather than producing a forced and potentially misleading single-class prediction.

The 73% accuracy of the Happy/Sad model, while lower than the Angry/Fear model, should be interpreted in the context of the task difficulty rather than as a limitation of the architecture. Happy and Sad drawings by children are inherently more similar visually than Angry and Fear drawings — both may contain similar figure sizes, similar stroke densities, and similar use of space. The 73% result is consistent with findings in the broader literature on children's drawing emotion classification, where inter-rater agreement among human experts on the same task is often only marginally higher. The hierarchical system's ability to compensate for this uncertainty through the Not Sure and blended outputs makes the overall system more honest and clinically appropriate than a system that always forces a definitive single-label answer.

7. PROJECT CONCLUSION AND FUTURE WORK:

7.1 Introduction:

This chapter concludes the DrawMind AI project report by reflecting on the outcomes of the development and evaluation phases relative to the objectives stated in Chapter One. The system was designed to provide a preliminary emotional analysis of children's drawings through a hierarchical deep learning pipeline, an explainability layer, and a natural-language interpretation interface accessible to parents and educators. The sections below assess the primary technical strengths of the realized system, acknowledge its current limitations with specificity, and propose concrete directions for future development.

7.2 Overall Strengths:

The DrawMind AI system demonstrated several notable strengths across its technical and functional components:

Hierarchical architecture that directly addresses the data scarcity problem: The most significant technical contribution of this project is the decision to decompose the four-class classification problem into two specialized binary models rather than attempting to train a single unified classifier on imbalanced data. The Happy/Sad model trained on the KIDO dataset (10,860 images) and the Angry/Fear model trained on a carefully deduplicated combined dataset (910 images) each operate on balanced, appropriately scoped training sets. This decomposition produced a system in which the Angry/Fear model achieves 93.41% test accuracy under Test-Time Augmentation — a result that would not have been achievable with a single four-class model trained on data dominated by the majority classes.

Rigorous dataset engineering: The MD5-based deduplication step applied during the construction of the Angry/Fear training dataset identified and removed 123 exact duplicate images — approximately 11.9% of the merged pool — that were present across two independently curated sources. This step was essential to prevent data contamination between the training and test sets and to ensure that the reported evaluation metrics reflect genuine generalization performance rather than memorization.

Custom preprocessing pipeline tailored to the drawing domain: The Master Filter — combining bilateral noise suppression, Laplacian sharpening, and MobileNetV2 normalization — was designed specifically to address the domain mismatch between children's drawings and the natural image domain on which MobileNetV2 was pretrained. This pipeline preserves stroke-level edge structure while reducing background noise, directly improving the quality of the features available to the model.

Multi-layer explainability: The combination of Grad-CAM spatial heatmaps and quantitative visual feature extraction (brightness, contrast, edge density, spatial brightness distribution) provides complementary views of the model's decision process. This multi-layer approach ensures that the system's outputs can be interrogated from multiple angles, which is important in a context where the end-user — a parent or educator — needs to understand and trust the analysis.

Natural-language report generation via Groq LLM: By translating structured model outputs and visual features into a structured JSON report through a carefully engineered prompt, the system bridges the gap between raw classification output and actionable guidance. The report distinguishes between emotional sub-categories, lists positive and concern indicators, and provides practical suggestions, all while explicitly framing the output as a preliminary analytical indicator rather than a clinical assessment.

Stable and scalable backend architecture: The FastAPI-based backend handles image upload, preprocessing, model inference, and database operations asynchronously, preventing blocking under concurrent load. The PostgreSQL schema enforces referential integrity through foreign key constraints across all six tables, ensuring data consistency as the user base scales.

7.3 Overall Weaknesses:

The system also has limitations that should be acknowledged honestly:

Moderate accuracy of the Happy/Sad model: The Happy/Sad binary model achieves 73% test accuracy under TTA — a result that, while honest, represents a meaningful error rate in a context where misclassification could have real consequences for a child's wellbeing assessment. This limitation is attributable primarily to the genuine visual ambiguity between happiness and sadness in children's drawings rather than to architectural shortcomings, but it nonetheless constrains the reliability of the complete four-class system for the positive/negative valence distinction.

Small test set for the Angry/Fear model: The Angry/Fear model's test set contains only 91 images (45 Angry, 46 Fear), which limits the statistical reliability of the reported precision, recall, and F1 metrics. Performance on individual class boundaries may be more variable than the reported aggregate metrics suggest, and the model's behavior on edge cases is not fully characterized by this evaluation.

Image quality sensitivity: The visual analysis pipeline is sensitive to the quality of the photograph submitted by the user. Poor lighting, shadows, oblique capture angles, or camera motion blur can degrade the preprocessing pipeline's ability to isolate stroke structure, affecting both model prediction and visual feature extraction. The system currently provides no image quality feedback to the user prior to analysis.

Dependency on external API availability: The natural-language report generation component depends on the Groq API, which introduces an external availability dependency. If the API is unavailable or rate-limited, the interpretation layer of the system is non-functional, degrading the user-facing output to raw classification results without explanation.

No longitudinal tracking: Each drawing is currently analyzed as a standalone input. The database schema stores historical results but the application does not yet implement trend analysis or visualization of a child's emotional profile over time. The absence of longitudinal analysis limits the system's utility for monitoring psychological development, which is one of the most clinically meaningful applications of repeated drawing assessment.

Limited dialect support in text input: The text input component processes parent-provided notes in standard Arabic and English. Regional dialectal variation — such as Levantine or Jordanian colloquial Arabic — may reduce the accuracy and nuance of the LLM's interpretation when the parent's input deviates significantly from formal standard Arabic.

7.4 Future Work:

Several directions are proposed for future development of the DrawMind AI platform:

Expansion of the Angry/Fear training dataset: The most immediate priority for improving system performance is the collection of additional labeled children's drawings in the anger and fear categories from culturally diverse sources. A training set of several thousand images per class would allow deeper fine-tuning of the backbone and improve the reliability of the hierarchical decision-logic module's four-class predictions. Collaboration with child psychology research institutions or educational organizations could facilitate the ethical collection of such data with appropriate consent procedures.

Longitudinal emotional tracking and trend analysis: Redesigning the application's analysis layer to aggregate drawing results over time and generate trend visualizations would substantially increase the system's clinical utility. A parent or teacher would be able to observe whether a child's emotional indicators are improving, stable, or deteriorating over a defined period, enabling earlier and more targeted intervention. This would require extending the database schema with timestamp-indexed analysis records and implementing a charting interface in the mobile application.

On-device model inference: Converting the trained models to TensorFlow Lite format for on-device inference would eliminate the latency introduced by server-side prediction requests, reduce dependence on network availability, and strengthen user data privacy by keeping drawing images on the user's device. This is technically feasible for MobileNetV2-based architectures, which were specifically designed for mobile deployment.

Voice input as a third analytical modality: Children frequently verbalize their drawings during the creative process, and the linguistic and prosodic content of these vocalizations can carry emotional signals independent of the visual content. Integrating a voice recording feature that captures the child's spoken description and analyzes tone, pitch, and keyword content alongside the visual and text-based inputs would provide a third independent modality for the emotional assessment pipeline.

Culturally grounded prompt augmentation via RAG: The current LLM prompt provides general guidance applicable to any context. Augmenting the prompt with retrieved content from a curated knowledge base of Middle Eastern child psychology research — using a Retrieval-Augmented Generation pipeline — would ground the generated recommendations in culturally appropriate frameworks, improving the relevance and specificity of the advice provided to the system's target user population.

Clinical validation study: A formal comparison of the system's classifications against assessments made by licensed child psychologists on the same set of drawings would provide an empirically grounded estimate of the system's agreement with clinical judgment. Such a study would be necessary before the system could be recommended for use in any para-clinical setting and would constitute a meaningful contribution to the research literature on automated psychological drawing analysis.

References

- [1] S. Çiftçi, H. Koçakoğlu, H. Karci, N. Karaca, and B. Söylemez, "Emotion Analysis of Children's Drawings," IEEE Access, vol. 13, pp. 159730–159746, 2025.
- [2] Serdar Ciftci, "Kido Children Drawing Dataset," Kaggle Dataset.
Available : <https://www.kaggle.com/datasets/serdarciftci/kido-children-drawing-dataset/data>
- [3] Vishmi Perera,
"Children Drawings Dataset," Kaggle Dataset.
Available: <https://www.kaggle.com/datasets/vishmiperera/children-drawings>
- [4] Mango-Ovirk,
"Emotion in Drawings Dataset," Roboflow Universe.
Available: <https://universe.roboflow.com/mango-ovirk/emotion-in-drawings>